

Annex 1: Proposal Template

Administrative proposal form

1. General information

Horizon Europe Cluster and Destination: CLUSTER 1 Health; (main) Destinations: D3, D4, D5

Proposal title: *AI-powered Digital Twin for next-generation lung cancer care*

Acronym: *IDEA*

Duration in months: 24

Keywords: Digital Twin, Artificial Intelligence, Non-Small Cell Lung Cancer, Virtual Scanning, Virtual Treatments

Abstract

Imagine a future where personalized healthcare will provide every individual with the right type of care in the right way and at the right time using digital twins (DTs), which are virtual representations of patients' health status. IDEA aims to shape this future in cancer care supported by the fact that DTs have seen faster growth in some industries such as manufacturing, but healthcare has been slow to embrace them. Furthermore, despite recent advances in computational capabilities and artificial intelligence (AI), cancer is one of the leading and most prevalent chronic conditions today and, in particular, lung cancer has the highest morbidity and mortality rate worldwide. Hence, **IDEA's main goal is to develop lung cancer DTs combined with AI to provide enhanced diagnosis and treatment decision support.** To this end, this project has four objectives:

1. to generate radiological lung digital twins for virtual scanning, serving as input for analysis and modeling: this reduces the need for multiple scans, minimizes radiation exposure, and matches each patient to the best-performing imaging modality and most appropriate imaging protocol;
2. to create DTs of lung cancer for virtual treatments, serving as input for planning the treatment, for predicting toxicity, patient response, and prognosis by leveraging AI techniques and multimodal data;
3. to propose appropriate data interfaces to guarantee the information flow from the physical patient to the virtual twin, to ensure data security and trustworthiness of the AI techniques through explainable AI methods;
4. to investigate the ethical concerns and impact to patients when integrating DT technologies in the current cancer care stream.

Through these objectives, IDEA improves the accuracy and efficiency of lung cancer diagnosis and treatment, reduces radiation exposure and carbon footprint, and ensures the security and fairness of AI techniques.

2. Participants and Contacts

Research Units (R.U.) carrying out the proposed work

R.U.1 (Coordinator): Unit of Computer Systems and Bioinformatics (COSBI)

Person in charge of the proposal:

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Key person:

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R.U.3: Unit of Diagnostic Imaging (DI)

Key person:

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R.U.4: Unit of Bioethics and Humanities (BH)

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It is worth noting that this proposal is endorsed by the following external institution:

- Centro Diagnostico Italiano SpA (CDI), Bracco Group, Milan, Italy

3. Budget of the proposal

RU name	Direct personnel costs (A)	Other direct costs (B)	Direct costs of subcontracting (C)	Total estimated eligible costs (A+B+C)	Co-funding (D)	Grant requested (A+B+C-D)
COSBI	53.400 €	2.000 €	0 €	55.400 €	0 €	55.400 €
RO	16.000 €	2.200 €	0 €	18.200 €	0 €	18.200 €
DI	16.000 €	2.200 €	0 €	18.200 €	0 €	18.200 €
BH	6.000 €	2.200 €	0 €	8.200 €	0 €	8.200 €
Total	91.400 €	8.600 €	0 €	100.000 €	0 €	100.000 €

Cover Page

Title of proposal: IDEA: AI-powered Digital Twin for next-generation lung cancer care

List of participants

Participant n°	Research Unit Name
01	Unit of Computer Systems and Bioinformatics, Department of Engineering, Università Campus Bio-Medico di Roma
02	Unit of Radiation Oncology, Department of Medicine, Università Campus Bio-Medico di Roma
03	Unit of Diagnostic Imaging, Department of Medicine, Università Campus Bio-Medico di Roma
04	Unit of Bioethics and Humanities, Department of Medicine, Università Campus Bio-Medico di Roma

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1. Excellence

1.1 Objectives and ambition

1.1.1 Executive summary

IDEA delivers digital twins (DTs) for new care generation in non-small-cell lung cancer (LC). DTs unlock access to personalize treatments for patients, giving physicians assistive artificial intelligence (AI) based technology that helps simulate in the virtual world the disease outcome in response to different possible treatments from a range of medical data collected in the routine, and updating the simulation as the stream of data is collected during the cure. DTs also enrich disease representation by virtualizing the scanning, offering complimentary intra- or inter-modality views of the disease translated from the original acquisition. IDEA's DT acts as a unique, smart system going well beyond existing attempts at 5P cancer care.

DTs are precise, virtual copies of living or nonliving real-world entities that are revolutionizing every industry, underpinned by advances in areas such as the Internet of Things, AI, extended reality, and cloud computing [1]. The primary purpose of a DT is to enable better decision-making by providing a detailed and accurate understanding of the physical object or system.

While DTs have seen faster growth in some industries such as manufacturing, other industries especially in the service sector, such as healthcare, have been slow to embrace them. Advances in precision medicine, supported to a great extent by increased computational capabilities and developments in analytics, have provided significant benefits to clinical care, trying to impact cancer care because this is one of the leading and most prevalent chronic conditions today and, in particular, LC has the highest morbidity and mortality rate worldwide. Indeed, in Europe, more than 4 million new cases of cancer were estimated to have occurred in 2020 [2], whilst LC accounts for more than 3 hundred thousand new cases in the same period, 85% of which are non-small-cell lung cancers (NSCLCs) that are less sensitive to chemotherapy and radiation therapy than small cell LC. Despite that, only about 20%-30% of newly diagnosed LC cases can undergo surgery with curative intent. Furthermore, many have a high risk of recurrence (25%-70%) due to preoperative micrometastasis.

To date, analytics advances and applications in cancer have shown promising performance, but they failed to provide an overall superior, personalized, individualized focus around care delivery for most patients because they neglect a key factor, which is the multimodality of human beings. Indeed, the decision-making process in cancer cure requires integrating heterogeneous data and germane knowledge: while the use of clinical, radiomics and genomics have shown promising results [2], their association remains unclear [3].

In IDEA we produce DTs embodying AI algorithms for cancer care that have the potential to better support patients and clinicians ameliorating most, if not all, of the current challenges; namely, providing enhanced real-time clinical decision support to clinicians, enabling patients to better understand their treatment options and likely impacts, assisting in developing a more realistic expectation and appreciation of the treatment journey, and helping patients to make better-informed discussions with their clinicians. Given the pressing need to address these healthcare challenges, the cornerstone of our work is to make real **intelligent healthcare**, where medical technology meets AI to facilitate improving and integrating isolated individual systems into a unified healthcare infrastructure. This happens in IDEA through DTs for LC care, where we turn a range of medical data into a virtual copy of the patient communicating with the physical world, enabling physicians to study and forecast disease outcomes in response to different possible treatments (WP4). Our DT provides also virtual scanning to analyze the diseases with different realistic intra- and inter-modality views (WP4), where every patient can be matched to the best-performing imaging resource and most appropriate imaging protocol. Our technology is **multimodal in the widest sense**, as we integrate many more data that are routinely used in the practice. We leverage **explainable AI (XAI)** techniques to give algorithmic insights to system designers, researchers, patients and healthcare personnel alike. We support varying stages of NSCLC, but also aim to **continually learn** to cope with other tumors. We will make our algorithms **open source** as we expect this approach to provide relevant advances in an area impacting a broad range of researchers, patients and health professionals. To achieve this goal, in addition to developing state-of-the-art algorithms (WP4), IDEA will require and generate complete **training data** sets through an elaborate data collection effort (WP3). To ensure a tight coupling between our specific goals, our target population's needs, and the ensuing results, this project will undertake an extensive **needs assessment** involving users and technology developers (WP1). We will also carry out a **rigorous evaluation** (WP5) to assess algorithms performance and system usability. All the results will be disseminated and exploited via a thorough **dissemination plan** (WP6).

Table 1.1 List of acronyms used in this proposal

Acronym	Definition	Acronym	Definition	Acronym	Definition
AI	Artificial Intelligence	DRL	Deep Reinforcement Learning	MDL	Multimodal Deep Learning
BERT	Bidirectional Encoder Representations from Transformers	DT	Digital Twin	MRI	Magnetic Resonance Imaging
CNN	Convolutional Neural Network	EHR	Electronic Health Records	NSCLC	Non-Small-Cell Lung Cancer
CT	Computed Tomography	GAN	Generative Adversarial Network	OS	Overall Survival
DL	Deep Learning	GDPR	General Data Protection Regulation	PET	Position Emission Tomography
DNN	Deep Neural Networks	LC	Lung Cancer	TRL	Technology Readiness Level
				XAI	eXplainable Artificial Intelligence

1.1.2 Objectives

Main Objective: Develop LC DTs combined with AI to provide enhanced diagnosis and treatment decision support, as well as a more individualized and patient-centered focus; that is, a balance between precision and personalization.

This general goal will be pursued by the following objectives merging the physical and virtual worlds:

Objective 1: To develop high throughput methods to generate radiological lung DTs for virtual scanning, serving as input for analysis and modeling.

The DTs will integrate AI models to offer intra- and inter-modality views: by translating inside a modality or between different medical imaging modalities (e.g., from computed tomography (CT) to positron emission tomography (PET), from CT to magnetic resonance imaging (MRI)), clinicians can virtually try other modalities and have an immediate insight into other information useful to deepen the status of the disease. Clinicians can also try other acquisition parameters for the same image modality, which can be mostly useful in MRI. This reduces the need for multiple scans, minimizes radiation exposure, and matches each patient to the best-performing imaging resource and most appropriate imaging protocol. It improves the sustainability by reducing the carbon footprint of medical imaging and by shortening waiting lists. DTs will be validated with NSCLC data from clinical sites, either publicly available or gathered from CDI. This will ensure system's accuracy, robustness, and generalizability. Use cases, developed with radiology and radiation therapy experts, will evaluate the efficacy of DT and AI in generating virtual data for personalized NSCLC treatments.

Type	Target	KPI
Technical	Data acquisition time: the time it takes to generate a DT of a patient's thorax should not surpass the time it takes a real scanner. This ensures that the process is fast and efficient, reducing patient wait times and improving workflow for healthcare professionals.	Acquisition time: <10 minutes per scan
	Image quality: the generated images should have a satisfactory signal-to-noise ratio. This ensures that they are of high quality and suitable for accurate analysis and diagnosis.	Signal-to-noise ratio of >25 dB for all the images.
	Image resolution: the generated images should have a high spatial resolution. This ensures that they are highly detailed and suitable for accurate analysis and diagnosis.	Spatial resolution of <0.5 mm for all the images.
	Image-to-image translation performance: the virtual scanner should provide faithful images, ensuring that the system can accurately generate images from multiple modalities, reducing the need for multiple scans and minimizing radiation exposure.	Peak signal-to-noise ratio >40 dB, Visual Turing test accuracy >0.8
	Modality-specific features: during modality conversion, the system should preserve modality-specific features, such as contrast enhancement, tissue attenuation, and noise characteristics. This ensures that the resulting images accurately represent the original data and can be used for accurate diagnosis and treatment planning.	No statistical difference in the modality-specific features before and after conversion.
	Modality flexibility: the system should be able to generate images from multiple modalities, including CT, MRI, and PET. This ensures that the system is flexible and can be used with a variety of imaging modalities, improving its overall usefulness and utility in clinical settings.	Generate images to at least 3 modalities.
	Reduction in energy consumption: by using AI techniques to reduce the need for multiple scans should bring a reduction of the energy required for scanning and processing images, leading to a more energy-efficient and environmentally friendly system.	Estimation of >10% reduction in energy efficiency ratio compared to current practices.
Clinical	Reduction in staging time: the time it takes to stage NSCLC should be reduced. This ensures that patients receive timely treatment, improving their chances of survival.	Reduction in staging time: >50%

	Radioprotection: the radiation exposure during the staging of NSCLC should be reduced. This ensures that patients are exposed to less radiation, reducing their risk of radiation-related complications.	Reduction in radiation exposure: >50%
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Objective 2: To create DTs of NSCLC for virtual treatments, serving as input for planning the treatment, for predicting toxicity, patient response, and prognosis by leveraging AI techniques and multimodal data.

IDEA implements semi-seamless connection and multimodal data exchange allowing the continuous, or at least periodic, twinning of the digital to the physical, thus moving a step forward with respect to the virtual and simulated models. DTs will embrace new AI-based models that, exploiting the most recent advances in multimodal deep learning (MDL) and deep reinforcement learning (DRL), have shown the possibility to integrate heterogeneous data in shared automatic representations. AI models help assess DTs' response to different dose distributions and fractionation, predict toxicity given by radiation treatment planning, and predict patient prognosis and response to therapy. Such representations can be updated every time a new bunch of data is collected. DTs will be tested and validated using real-world NSCLC data from multiple clinical sites, already available in public repositories or collected from CDI. This will ensure that the system is robust, accurate, and generalizable across different patient populations and imaging modalities.

Type	Target	KPI
Technical	Data integration: the system should be able to integrate multiple sources of heterogeneous data, including medical imaging, genomics, electronic health records (EHRs) and clinical data, into a single DT model. This ensures that the resulting model is comprehensive and can be used to inform treatment decisions accurately.	The number of integrated modalities should be at least 3.
	Semi-seamless data exchange: the system should enable near-real-time data exchange between the physical and DTs to ensure the most up-to-date information is used for treatment planning and prediction. This ensures that treatment decisions are based on the most current data available.	Enable automatic data retrieval from medical data repository.
	AI model accuracy: the AI models developed should accurately predict patient response and prognosis based on the DT data. This ensures that the predictions are useful and reliable for clinical decision-making.	Ct-index >0.8.
	Model interpretability: the AI models should be interpretable and explainable to clinicians, so they can understand the basis for the predictions and make informed treatment decisions. This ensures that clinicians can trust and use the predictions in their decision-making process.	Verify with clinical assessment tests which all result in positive feedback.
	Validation: DTs should be tested and validated using real-world NSCLC data from multiple clinical sites to ensure its robustness and generalizability across different patient populations and imaging modalities. This ensures that the system is reliable and useful in clinical settings.	Test and validate on at least 2 external datasets.
Clinical	Reduction in treatment planning time: the time it takes to generate personalized treatment plans for NSCLC should be reduced. This ensures that patients receive timely and individualized treatment plans, improving their chances of survival.	Reduction in treatment planning time: >50%
	Treatment planning safety: DT should lower treatment planning toxicity compared to conventional planning methods. This ensures that patients receive the most effective and less toxic treatment possible.	Reduce treatment planning toxicity >10%
	Survival rate prediction: The system should result in increasing the possibility to predict the survival rate of patients with NSCLC, in comparison with standard prediction based on clinical parameters. This ensures that the system is effective in tailoring patients' treatment according to different prognoses.	Increase in survival rate prediction: >15%

Objective 3: To propose appropriate data interfaces to accommodate DTs in a healthcare development environment.

The project studies interfaces to contain and guarantee the information flow from the real to the virtual space, i.e., the data flow from the physical patient to the virtual twin. Such interfaces are the enabler of data exchange, allowing the convergence and synchronization of the virtual and physical systems. DTs need to ensure not only data security, but also the trustworthiness and fairness of the AI techniques, which we will pursue through XAI methods. The software tools will be made available by integration with existing open-source healthcare platforms by means of user-friendly interfaces for clinical use.

Type	Target	KPI
Technical	Interface integration: successfully integrate the DTs and associated AI models into existing open-source healthcare platforms with minimal technical difficulties.	100% successful integration within 3 months of completion of development.
	Trustworthiness and fairness: develop and implement appropriate measures to ensure the AI models used in the DT are trustworthy, unbiased and fair.	Improve Infidelity and Sensitivity by at least 10% compared to state-of-the-art XAI techniques.

Clinical	Clinical adoption: start with a minimum adoption rate of 50% of healthcare clinical practice within 6 months of implementation.	50% adoption rate within 6 months.
	User satisfaction: conduct user satisfaction surveys based on the System Usability Scale to assess the satisfaction level of healthcare providers using the DT interfaces including XAI feedback.	To achieve at least 80 on the System Usability score among users.

Objective 4: To investigate the ethical concerns and impact on patients of DTs in cancer care.

While a DT in NSCLC treatment can benefit the cost and time savings greatly because its services produce a lifetime growth for the hospitals and organizations, there are many ethical concerns to integrate DT technologies in the current cancer care stream. There is also the need to address the impact on patients' safety.

Type	Target	KPI
Ethical	The following ethical items will be assessed: Benefit-harm balance, Autonomy, Respect for persons, Justice and Equity, Legislation, Ethical consequences of the HTA; The following ethical items will be assessed: Patient safety, Occupational safety, Environmental safety, Safety risk management	Implementing or refraining from the implementation of the technology
Clinical	Improvement in physicians' satisfaction: conduct surveys based on the System Usability Scale to ensure that physicians are satisfied with the quality of care they deliver.	Improvement in physicians' satisfaction by at least >10% on the System Usability score.
	Clinical adoption: The DT should be adopted and used by multiple clinical Units. This ensures that the system is widely adopted and used in clinical practice, increasing its impact on patient care.	Used by at least 2 clinical Units within 2 years of its implementation.

1.1.1 Alignment with the Horizon Pillars

This project meets all destinations of the Horizon Europe Cluster 1 "Health" call, as described in Table 1.2.

Table 1.2. Alignment of IDEA with the Horizon Pillar

Destination	Project Coherence and Motivations
D1: Staying healthy in a rapidly changing society	The development of a LC DT with AI will contribute to disease prevention and promote healthier lifestyles. By using AI techniques to analyze data, the DT can provide personalized diagnosis and treatment recommendations, which will ultimately improve health outcomes for patients.
D2: Living and working in a health-promoting environment	The project will incorporate various data modalities, including radiological imaging and medical records, to generate a comprehensive DT of the patient. By understanding these determinants of health, the project can develop personalized and effective treatment plans for patients.
D3: Tackling diseases and reducing disease burden	The use of AI techniques in the analysis of patient data will provide healthcare providers with a better understanding of LC, enabling them to develop more effective treatment plans. The DT technology can also be applied to other diseases, increasing its impact on disease burden reduction.
D4: Ensuring access to innovative, sustainable and high-quality health care	By using AI techniques to analyze patient data, the project aims to develop personalized and cost-effective treatment plans for patients. The development of a LC DT could also help reduce healthcare disparities by providing access to state-of-the-art diagnostic and treatment methods to all patients.
D5: Unlocking the full potential of new tools, technologies and digital solutions for a healthy society	The project will focus on the development of appropriate data interfaces for the DTs, ensuring the secure and ethical integration of this technology into healthcare systems.
D6: Maintaining an innovative, sustainable and globally competitive health-related industry	The development of a LC DT with AI is a breakthrough technology that could position the EU health industry as a leader in cancer diagnosis and treatment. By leveraging AI techniques and multimodal data, the project can develop a more innovative and sustainable healthcare system.

1.1.2 Beyond the state-of-the-art and Ambition

The proposed project will push the boundaries of the state-of-the-art in NSCLC care by offering DT that pairs with cutting-edge technologies in AI, including image generation, multimodal and reinforcement learning, as well as XAI. This would offer unlimited possibilities to the traditional approach of "first treat and then adapt", "first verify and then learn" as the project provides the opportunity to merge the virtual and physical worlds and pave the way for personalized and precise treatments of NSCLC, a chronic disease that impacts patients'

life and the European healthcare system. Furthermore, this project will offer the possibility for European healthcare and research to be competitive without compromising quality for patients.

Pioneering the role of DT in cancer care is the main ambition of this project. DTs, succinctly described as the digital representation of a physical object, is a concept that has emerged relatively recently with increasing application in the manufacturing industry [4], [5]. The application of this concept to the healthcare domain stems from past research on modeling and simulations. However, the difference between the conventional simulation and the DT is broadly a matter of scale: while a simulation generally studies one distinct process, a DT can itself simulate any number of variables to study multiple processes. DT research is in an early stage despite its potential to provide enhanced clinical decision support and enable more patient-centric, and simultaneously more precise and individualized care to ensue. Existing attempts to deliver DTs have focused on the heart [6], airways system and its fluid dynamics [7], blood vessels for aneurysm treatment [8] and cancer [4], [9]. In particular, they are the only works that, to the best of our knowledge, have proposed a DT in cancer care; in [4] the authors focused on AI-driven decision support in endometrial cancer and in our work [9] we developed a DRL controller to optimize LC radiation therapy by using patient-specific CT scans and a virtual radiotherapy environment, which can adapt to different tissue aggressiveness and outperforms the current clinical practice of uniform dose delivery. On the one side, such works have pushed forward the research in this field but, on the other side, they suggested to us several directions of investigations that needed to be addressed that inspired our ground-breaking objectives. To pursue them, our proposal addresses the following ambitious concepts, issues and problems, which are tackled by the appropriate and cutting-edge approaches and methods described in section 1.2.

Radiological DT exploits AI-based image generation for virtual scanning. While image generation for modality translation has made significant progress in recent years [10] thanks to advances in deep learning (DL) and its subfield that studies generative models, there are still several limitations to the state-of-the-art techniques. This project addresses the main ones, namely:

- **Limited diversity:** While DL models can generate high-quality images, they often lack diversity, i.e., they tend to generate similar images from different modalities with minor variations [10]. This reflects in models that collapse and cannot reflect the human and the scanner variability.
- **Lack of interpretability:** Some image generation techniques may produce high-quality images, but it can be challenging to interpret the internal representations that the model has learned [10]. This can make it difficult to understand how the model is generating the images and to diagnose any errors or biases that may be present [11].
- **Image translation for applications to DTs:** Generative approaches proposed to date have been designed for offline applications, where there is no exchange of data between the physical and the virtual world. This reflects in extremely time-consuming procedures to train the DL models that make small throughput of these approaches.
- **Clinical validation:** While image generation techniques may generate realistic images, there is a need for rigorous clinical validation to ensure that they are accurate and reliable in clinical settings [10]. This requires careful evaluation and testing, which can be time-consuming and expensive [11].

NSCLC DT for treatment planning and outcome prediction. Our experience of the world is multimodal, and we say that a problem is multimodal when it includes multiple modalities. Our hypothesis is that multimodality is an enabling factor to develop the DT of NSCLC and to leverage AI models for decision support. Despite there being a general consensus on the importance of mining together multimodal quantitative biomarkers, such as those expressed by radiomics, digital pathology, physio-psychological data, and EHRs, the literature has attempted to mine only pairs of such modalities [12]–[14]. Furthermore, patients' picture evolves over time, e.g. the size of the tumor changes during the treatment: there is therefore the need for allowing periodic twinning of the physical and virtual world. The main ground-breaking directions of investigation are:

- **Resilience of learning paradigms:** the use of data in-the-wild opens unprecedented challenges to AI models. Indeed, whole successful training of AI models usually requires a large amount of diverse data [5], data scarcity is a major limitation in healthcare due to issues such as data privacy, lack of interoperability, and data quality, resulting in models overfitting. There are further difficulties caused by missing data and modalities that harm the robustness of AI models.
- **Transparent and trustworthy decisions:** Applying DTs in combination with AI to complex NSCLC contexts would assist clinical decision-making: despite we can continuously search for better performing models, the opacity of AI models could limit physicians', patients' and caregivers' trust in the decisions, and this is even more urgent in multimodal learning where understanding the way a model combines the heterogeneous source of information is not trivial [15], [16].

Data interfaces and storage for DTs. There are three types of communication processes that need to be designed: i) between the physical and the virtual twin, ii) between the DT and domain experts, which interact and operate on the DT, through usable and accessible interfaces, iii) between the DT and different DTs in the

surrounding environment. All the exchanged data must be stored in a data storage system, but they also need to be transmitted and processed so that they are conveyed back and forth to the DT: this asks hospitals and organizations to ensure that the patient information is secure and trusted throughout the whole DT process[17]. In terms of data interfaces, there are a number of protocols and formats that are commonly used for exchanging data between different components of a DT. These include:

- **OPC-UA (Open Platform Communications - Unified Architecture):** a standard protocol for industrial automation that provides secure and reliable communication between devices, machines, and systems.
- **MQTT (Message Queuing Telemetry Transport):** a lightweight messaging protocol designed for use in IoT devices, which provides a reliable and efficient way to exchange data between devices.
- **REST (Representational State Transfer):** an architectural style for web services that provides a simple and lightweight way to access and exchange data between applications.

In terms of storage, among the different approaches available, for data storage we will exploit NoSQL databases, i.e., state-of-the-art non-relational databases that are designed to handle unstructured and semi-structured data, such as JSON or XML. We will also guarantee cryptography, authentication and authorization.

Radiation therapy DT for treatment planning and patient outcome. Radiation therapy is one of the main approaches to LC treatment, and its planning is crucial for therapy outcome. However, the current practice that uniformly delivers the dose does not take into account the patient-specific tumor features that may affect treatment success. Since radiation therapy is by its very nature a sequential procedure, DT can effectively interact with the physical world to get information on the tumor evolution as well as on patient health status to support therapy planning. Furthermore, short, medium and long-term prediction of the outcome is still an open issue addressed in the literature considering, in most cases, only unimodal data [12]–[14]. The few works that try to mine together multimodal data in the large majority of the cases tackle a classification task [12]–[14], therefore fixing the time and thus neglecting to predict the survival at different time stamps [12]–[14]. In this respect, this work moves towards the following direction of innovations:

- **Twining radiation therapy planning and toxicity:** to deliver personalized treatments there is the need to optimize the daily dose fraction delivered to the patient as well as to forecast induced toxicity in the tissues. Both issues ask for twinning the tissue radiosensitivity and disease response by means of data collected during the treatment itself.
- **Twining outcome prediction:** When defining the radiation therapy planning the experts would benefit not only from the prediction of the outcome but also from the chance to visually assess how the tumor would likely evolve over time.

Ethical concerns and impact on patients of DT in cancer care. Exploiting DTs in cancer care is still a research problem; nevertheless, it is important to investigate the different directions of the impact of these technologies. Our innovations will be directed towards the following issues:

- **Validation and Regulatory challenges:** Evaluation of DT in cancer care is in an early stage. Leveraging methods to estimate the performance of AI models, we plan to extend them for DT assessment in clinical settings, not only considering the prediction rate but also the impact on patients' treatment, explainability, etc. These issues will be studied also taking into consideration that AI models used in healthcare must comply with strict regulatory standards [18], such as the Health Insurance Portability and Accountability Act and the General Data Protection Regulation (GDPR).
- **Ethical assessment and social impact:** Emerging/convergent technologies, DT and AI included, are a new field of techno-science characterized by the synergetic integration of several previously separate scientific fields, envisaged as a broad innovation that will lead to a radical modification of human beings and humanity itself [41]. Ethical and bio-legal reflections show significant critical issues concerning the substantive quality of content and data (big bad data), authentication of information, transparency of algorithms, affordability and access without discrimination to such new technologies, the need to balance the protection of fundamental individual rights (privacy and autonomy) with the rights of the community (collection and use), and the possibility that health data collected by technology providers may exceed the required purposes (behavioral data surplus), and the definition of liability for malpractice [41]
- **Patient safety assessment:** Technological developments call for a reflection on the regulation of liability arising from the use of AI [42]. The focus of the matter concerns the level of autonomy of self-learning machines and the degree of potential human control over choices and behavior [43]. In this setting, WHO had also emphasized the need for analysis of the likelihood of an adverse event, which depends on the occurrence of "latent deficiencies" in the system, i.e., inadequacies or errors in design, organization and control, which remain silent in the system until a triggering factor makes them manifest in all their potential, causing more or less serious damage [44].

1.1.3 Innovation potential

The proposed project represents significant innovation potential in the field of cancer care. The development of a DT for radiology and radiotherapy, leveraging AI techniques that exploit MDL integrated with XAI methods are novel concept with the potential to transform the way we plan treatments. Currently, the only theoretical example of a DT in healthcare exists in the field of cardiology [19], but there have been no actual implementations yet [20]. Furthermore, few entities are committed to advancing radiology in the DT scenario [13]. This project first pairs different image modalities offering a virtual comparative scanning where radiologists and radiotherapists can easily assess different imaging techniques and test different acquisitions on the patient's virtual twin. The project twins physical and virtual patients suffering from NSCLC, providing a system that enables a semi-seamless exchange of data collected during the therapy: this would allow investigating how the tumor will respond, simulate and predict tumor evolution over time and, definitely, test different decisions on the DT to find out the best-personalized therapy. As nothing really exists further to position papers [5], we deem that such a project represents a significant innovation compared to existing products and services on the market.

We also carried out a **preliminary patent search**: the results indicate that the proposed system's unique combination of features and methods has the potential to be patented. Further investigation into the intellectual property landscape will be undertaken as the project progresses to ensure that the developed innovations are protected and that they can be commercially exploited.

We also carried out a research project search on the CORDIS platform, as described in section 1.2.3.

1.1.4 R&I Maturity and preliminary results

The proposed work on the development of a DT for radiology and radiotherapy coupled with AI is currently positioned in the early-to-middle stages of the R&I maturity spectrum.

IDEA starts at Technology Readiness Level (TRL) 2 and moves the technology up to TRL 5 at the end of the project. At the beginning of the project, the research will primarily focus on the development and optimization of the individual components, such as the image translation techniques, and AI resilience into-the-wild through an advanced approach for MDL and XAI. This corresponds to a starting TRL equal to 2, indicating that the project is at the stage of technology concept formulation.

As the project progresses, the individual components will be integrated, and the DT system will be iteratively refined, tested, and validated using real-world data from multiple datasets, many publicly available and two private. This will advance the project along the R&I maturity spectrum, moving toward the development of a functional prototype. By the end of the project, the DT system is expected to reach TRL 5 or 6, signifying that the technology has been demonstrated in a relevant environment.

Several consortium partners have collected experience moving from TRL2 to TRL5

The development of a user-friendly interface for the DTs will facilitate their adoption by healthcare professionals and contribute to the project's progress toward market readiness. After the completion of the project, further steps such as additional validation, regulatory approval, and commercialization efforts will be necessary to fully transition the technology from lab to market.

However, while DTs are intelligent systems, it is important to understand that they are not always fully autonomous. Indeed, AI-based applications and DTs require a lot of human intervention, particularly in scenarios where they are exploited to provide answers such as diagnosis and treatments. DT's AI instead of human intelligence is not necessarily more efficient; however, human skills boosted by DTs' analyses, predictions, and recommendations are undoubtedly more productive.

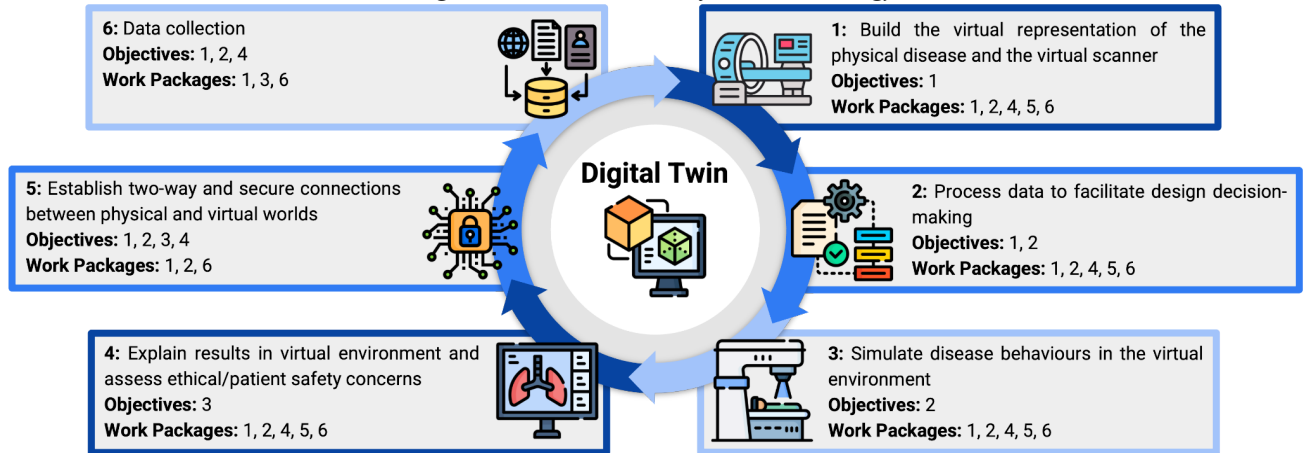
Table 1.3. Preliminary data and results that can be used in the project.

RU	Preliminary data and results that can be used in the project
COSBI, RO, DI	We account for data collected from >350 patients included in the CLARO dataset [13], which will be included in the retrospective cohort.
COSBI, RO, DI	We possess extensive expertise in mining radiomics and pathomics data for NSCLC, as we have explored AI-based methods to predict the overall survival (OS) and the treatment response of stage III and IV patients [12], [14]. We have also investigated the use of clinical name-entity-recognition to extract relevant clinical information from Italian-language EHRs.
COSBI, RO	We present a novel approach for handling missing data in the analysis of OS for NSCLC patients, utilizing clinical data from the CLARO project [13]. Our approach overcomes the limitations of traditional imputation methods, reduces bias, and improves the accuracy of the final analysis.

RO	Two awards. In 2019, we received the FONICAP/AIC-NET award for young researcher clinical study, and in 2018, we were honored with the Guido Berlucchi Foundation Best Oral Presentation Prize.
COSBI	We have proposed a novel approach for evaluating the quality of 2D CT image generation using the Wasserstein generative adversarial network (GAN) framework. This methodology was tested on the CLARO dataset, yielding promising results.
COSBI, DI	In 2022-2023, we have developed and validated DL models for Virtual Contrast Enhancement in Contrast Enhanced Spectral Mammography, which enables the generation of virtually recombined images using only the corresponding Low-energy image as input. The goal of this approach is to enhance contrast without the need for additional high-energy images.
COSBI, CDI	We have conducted a study using a radiomic-like approach to investigate whether AI can process chest X-rays and clinical data to identify COVID-19 patients at an early stage who are at risk of severe outcomes. Our study employs MDL with early, joint, and late fusion to optimize the understanding of "when," "which," and "how" to fuse the different modalities [21]–[23].
COSBI, CDI	We studied multimodal data integration in radiomics for Acoustic Neuroma and Breast Cancer [45].
COSBI	In March 2022, we were awarded first place in the Covid CXR International Hackathon (ELLIS, Dubai Expo) by developing a novel MDL approach that integrates chest X-rays and clinical data. Our approach is designed to be intrinsically explainable, providing greater interpretability and transparency.
COSBI	In December 2021, we were awarded first place in the “AI against COVID-19: Screening X-ray Images for COVID-19 Infection”, IEEE SIGHT, University of Waterloo, by developing a novel ensemble approach that integrates multiple networks trained on chest X-rays. Our approach is designed to be intrinsically explainable, providing greater interpretability and transparency.
COSBI	We have developed and validated DL models for MR-to-CT image translation, enabling the generation of synthetic CT images from real MR images. Our approach integrates novel data augmentation techniques that enhance the generalization and robustness of the networks.
COSBI	In the context of synthetic CT generation, we have developed denoising techniques to reduce the radiation dose while maintaining the quality of the virtual scanner provided to the clinician. By introducing a novel loss function, we have achieved superior denoising quality and faster training.
COSBI	We developed a DRL-based controller to personalize radiation therapy treatment, demonstrating superior performance compared to the current clinical practice. A virtual radiotherapy environment estimates the controller parameters using Particle Swarm Optimization, utilizing routinely collected CT scans.

1.2 Methodology

Figure 1. Schematic view of the Methodology



Although there are currently various DT definitions, the most used is “DT means an integrated multiphysics, multiscale, probabilistic simulation of a complex system, which functions to mirror the life of its corresponding twin”. DTs consist of three parts: physical entities, virtual models and the linkage between physical and virtual worlds [10]. The *physical entities* are the real patients’ lungs with NSCLC, which have different characteristics, behaviors, and outcomes in the course of the disease, and that generate a lot of data during the diagnosis and the treatment. The *virtual models* are images and mapping of the physical disease in the virtual space. They could reflect the whole lifecycle of the disease, as well as simulate, monitor, diagnose, predict, and state the behaviors of the physical entities. The virtual models include not only anatomical models but also pathological information that can be inferred from the images, and other data properties collected in semantic information (e.g., lung functions, staging, risk factors, etc.). The *connected data* include the subsets

of physical data and virtual data, as well as some “new data” that are acquired after the integration, fusion, and analysis of physical data and virtual data. Through screening medical examinations, the disease attributes, status, and other data are fed back to the virtual models, whilst the simulations on the virtual twins support the decision on the patient treatment, achieving a two-way data transmission process.

We design the DT to continuously collect data and knowledge on the entire lifecycle process of the disease, such as treatment planning, delivery, and outcome evaluation. And this data and knowledge could continue to be reused and improved.

1.2.1 Use cases

We offer use cases to illustrate how IDEA’s innovative DT improves the treatments and outcomes of people who are affected by NSCLC. They highlight the wide-ranging applicability of IDEA’s solution.

1.2.1.1 Person A: a patient suffering from NSCLC at stage III

We are now going through the diagnostic-therapeutic history of Person A, a 71-year-old man with stage III LC for which the DT is useful in several stages. He had a 30-pack year smoking history and presented with cough, decreased hemoglobin levels, and platelet counts. His CT scan reveals an irregular mass in the left upper lobe and a 2 cm subcarinal node. To better understand tumor behavior, and to plan the right treatment, the DT translates from Person A’s CT scan to 18FDG PET/CT: this would help his radiation oncologist immediately identify whether the lung mass and mediastinal nodes have pathological uptake, to boost the diagnostic process while reducing the radiation exposure. Subsequently, Person A undergoes a bronchoscopy, revealing a neoplasm at the opening of the upper branch of the left bronchus. Histology confirms squamous cell carcinoma, PDL 15%. After complete staging, the tumor is classified as clinical cT2bN2M0, stage IIIA, according to TNM classification of the Union of International Cancer Control (UICC), 8th edition. Person A is a candidate for definitive chemoradiation followed by immunotherapy. The DT visualizes different treatment plans with varying isodose distributions applied to the patient’s tumor, along with estimating the risk of pulmonary, heart, and esophageal radiation-induced side effects. This information would shorten treatment planning times and facilitate preventive strategies to reduce radiation-induced side effects. Before starting chemoradiation treatment, the DT provides information on tumor reduction during radiation therapy based on daily delivered dose using daily treatment images. Integrating this data with baseline and clinical data from EHR, his clinicians better evaluate the real dose to be delivered to the patient, maximizing the therapeutic ratio of treatment. Otherwise, the patient would undergo standard treatment with standard dose recommended for locally advanced disease. Finally, the DT integrates all information deriving from images, baseline features, and clinical data to predict the Person’s A OS: his clinicians use this information during the follow-ups to evaluate the possible outcomes and to tailor the treatment plan according to the Person’s A prognosis.

1.2.1.2 Person B: a patient suffering from NSCLC at stage IV

We are now going through the diagnostic-therapeutic history of Person B, a 64-year-old woman with stage IV LC for which the DT is useful in two stages. She had a 30-pack year smoking history, suffered from hypertension, and experienced persistent chest pain for several weeks. Her chest CT scan shows an 8 cm right upper lobe mass infiltrating the chest wall anteriorly and multiple enlarged mediastinal lymph nodes. Due to the advanced thoracic disease the patient underwent a total body CT scan, revealing suspicious brain lesions (at least 3), a lytic area of the right sacrum and a 3 cm liver hypodensity in the 4th segment. To better understand the nature of these suspicious lesions, the DT translates from Person B’s brain, liver and pelvic CT scan to the corresponding MRI, by better characterizing each lesion, adding functional features like diffusion and perfusion and additional information, such as the definitive number of lesions. Moreover, it would save procedure time and reduce the radiation exposure of the patient. Subsequently, Person B undergoes a CT-guided biopsy, revealing a lung adenocarcinoma, all wild type PDL 70%. The tumor is classified as stage IV, according to TNM classification 8th edition and Person B is a candidate to first line treatment with immunotherapy alone. At this stage, the DT merges all information deriving from images, baseline features, and clinical data to predict the Person’s B OS: her clinicians use this information during the follow-ups to evaluate the possible outcomes and to tailor the treatment plan according to the Person’s B prognosis.

1.2.2 DT-driven design

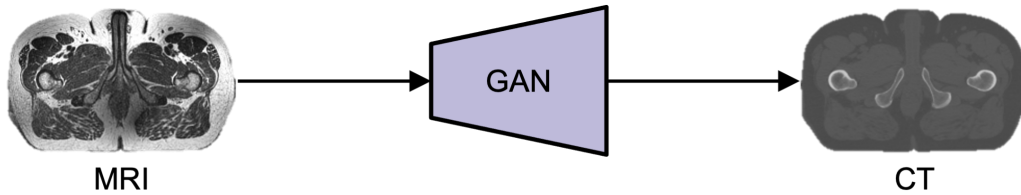
Figure 1 shows the main six steps to create our fully functional DT, which are explained below.

1.2.2.1 Build the virtual representation of the physical disease and the virtual scanner

The enabling technologies of this step are 3D visualization of medical images overlaid by other data collected during the therapeutic path. The virtual disease includes three aspects: elements, representations, and

rules. At the level of elements, the virtual model mainly includes the anatomical information of the disease, collected by imaging devices, as well as all the other information collected at the time of the diagnosis or during any follow-up (e.g., data stored in EHRs, semantic data). At the level of representations, the DT will make available a *virtual scanner*, which includes the virtual translation of one image modality into another one, such as from CT to MRI domain. In this respect, we define this task as an image-to-image translation problem [1] where the goal is to learn how to map images from a source domain to a target domain. To alleviate the complexity of data collection, we will use unpaired methodologies where we do not need well-paired datasets where each image must be present in both image domains. Further to well-known Variational Autoencoders, the developed model may be based on the CycleGAN model: this model learns a mapping between two image domains (A and B) by transforming from domain A to domain B, and simultaneously learning to transform from domain B to domain A [24]. This cycle-consistency allows domain-to-domain translations without manually pairing up samples in domains A and B. As for model training, we will use unpaired publicly available datasets (mentioned in section 1.2.2.6) and data provided by our project partners. Figure 2 shows an example of our preliminary results when using cycleGAN to translate from MRI to CT scans. The evaluation is based on qualitative and quantitative image quality assessment scores [18]: the former relies on human experts' evaluation (e.g., Turing test), and the latter exploits mathematical functions to objectively and reliably quantify the image quality, and many of them can be used using unpaired datasets (e.g., Frechet Inception Distance, Signal-to-noise-ratio) [2]. The test set will be composed of patients' scans extracted from the same repositories mentioned before, using hold-out validation. One of the main challenges we may encounter is the need for a large amount of high-quality data for the training and validation of AI models. To overcome it, we will leverage public datasets that are easily available in an unpaired case. Additionally, we will engage with experts in medical imaging and radiation therapy to ensure that the DTs and AI models are clinically relevant and aligned with current standards of care.

Figure 2. Example of virtual scanning (our preliminary results). Image-to-image translation from MRI to CT



1.2.2.2 Process data to facilitate design decision-making

Data collected from different sources (images, EHRs, semantic information) are analyzed, integrated and visualized, and used for decision-making. Representation learning enables us to convert data into more concrete information that can be directly queried by algorithms, through deep neural networks (DNNs) that have proven to succeed in this task. We have already preliminary results indicating that ensembles of convolutional neural networks (CNNs) and transformers are the net architecture that we plan to investigate. Possible strategies to merge the information are early and intermediate fusion. Early fusion joins different input modalities into a single vector (e.g. concatenation, pooling, or by applying a gated unit), leveraging on abstract representations extracted by modality-specific networks, such as CNNs for images and Bidirectional Encoder Representations from Transformers (BERT) networks for EHRs. Intermediate fusion allows us to define which, how, and when to join the different modalities. Indeed, we plan to join learned feature representations from intermediate layers of DNNs with features from other modalities as input to the final model. A first fusion strategy gradually concatenates the learned imaging features and the EHR features, taking into account the correlation among the modalities (e.g., by first fusing highly correlated modalities), the differences between the dimensionality and dynamic range of imaging and clinical features (e.g., by replicating and scaling the descriptors). A second fusion strategy uses DNNs as feature extractors, defining a single multimodal multi-task loss function using multiple input combinations. During the joint training process DNNs, we plan to use soft constraints to penalize differences between pairs of shared layers to enforce the network weights to be shared, also optimizing the consistency among network weights by a robust norm [26]. When formulating the loss function to train the DNNs, we enforce intermodality and intramodality relationships using information-theoretic regularizers and structured regularization [27] and introduce one or more regularization terms to get stability, algorithmic speed-up and to reduce overfitting.

With reference to data visualization, we integrate off-the-shelf technologies to present data in a more explicit fashion. Finally, advanced AI intelligence techniques can be incorporated to enhance a DT's cognitive ability

(e.g. reasoning, problem-solving and knowledge representation), so that certain relatively simple recommendations can be made automatically.

1.2.2.3 Simulate disease behaviors in the virtual environment

Here we simulate in the virtual world tumor behaviors, their evolution, and patient response to treatment. We simulate patient outcomes predicting the OS, which refers to the length of time from initial cancer diagnosis to the time of death. Identifying subgroups of patients who have a higher or lower chance of survival is critical in developing effective strategies to improve OS rates. To this end, the DT will be integrated with AI models tackling the survival task by considering that data is multimodal by its very nature, censored, heterogeneous, and contain missing information, and yielding time-varying conditional survival predictions, which can be used to generate patient survival curves for the covered time span. Methodologies will leverage deep networks, one per modality, and trained to generate modality-specific feature representations that are fused as explained in section 1.2.2.2. A final neural network takes the fused feature representation and outputs both a conditional survival probability for each of a set of pre-defined follow-up time intervals and the predicted survival curve. The validation of the AI models uses k-fold cross-validation and .632+ Bootstrap, which offer complementary insights into the results since the former provides less biased error estimation but with higher variance, while the latter estimates the error with lower variance. Our DT simulates the toxicity and the daily dose fraction delivered to the patient: while the current practice uniformly delivers the dose neglecting patient-specific tumor features, we will exploit DRL to transform this field by exploiting the intrinsic sequential procedure of radiation therapy. Our DT will simulate the patient-specific tissue radiosensitivity by combining both the effect of the radiotherapy treatment and cell growth, which is modeled by mixing ordinary differential equations with a data-driven approach where the equation's parameters are learned by an evolutionary algorithm that searches for the best solution. The space of DRL actions will be defined to fit the constraint of the maximum dose that can be daily delivered to each patient. The rewards function will be defined to model different treatment strategies that deliver different dose distributions during the treatment. The entire AI system will be fed by data collected at each follow-up (CT images, EHRs, semantic information), achieving a two-way data transmission process. The validation will compare the DRL policies and the standard fractionation scheme, exploiting a virtual population generated by the DT with different tissue radiosensitivities. We will qualitatively compare the average daily dose with the evolution of the tumor fraction of the initial volume in the fractionation time, and we will quantitatively convert the delivered total doses into their 2 Gy equivalents to compare AI-driven plans with those scheduled by the physicians. Furthermore, our DT will mirror treatment toxicity given by different treatment plans, so that clinicians will select the best plan not only according to dose spatial distribution and its impact on target organs, but also considering the likelihood of induced toxicity according to CTCAE v6.0. To this end, we will train and evaluate joint learning models using methodologies similar to those already described to predict the OS.

1.2.2.4 Explain results in virtual environment and assess ethical/patient safety concerns

The decisions suggested by AI models in the digital world need to be explained to increase the trust and fairness of the algorithms. To this end, we make the hypothesis that the recent advances in XAI can be translated into the multimodal AI-driven models developed. Saliency maps are data- and model-agnostic and can be applied to any modality, e.g., medical images [28]. Concept Activation Vectors are a promising human-centered technique providing an interpretation of a neural net's internal state in terms of human-friendly concepts, already applied in radiomics [29]. Attention mechanisms have been applied to EHRs [30] assigning more weight and importance to individual features that mostly contribute to the final prediction. In multimodal tasks, they play a crucial role in the alignment and fusion across the different modalities. We will include attention at different levels: (i) within the modality fusion, to evaluate the contribution of each modality to the final prediction; (ii) after embedding layers in EHR analysis, to check which EHR data mostly contribute to the final prediction; and (iii) between images and text embedding, to learn the alignment between image models and reports. To ensure the validity of the developed XAI methods, we will conduct a series of assessment tests aimed at evaluating their ability to provide clear and meaningful explanations for their predictions, also according to the evaluation metrics proposed in this context [31]. Additionally, we will assess the usefulness of these explanations to clinicians and evaluate the impact of the XAI method on clinical decision-making and patient outcomes to gauge its clinical relevance.

The ethical analysis will consider the prevailing social and moral norms and values relevant to the technology in question. This will involve understanding the consequences of implementing or not implementing a healthcare technology in two respects: with respect to prevailing social values and with respect to the norms

and values that the technology itself constructs when it is put into use. In addition, the ethical evaluation will address the ethical consequences of the choice of specific endpoints and the possible existence of ethical problems associated with the economic evaluation. Ethical and deontological framing will be carried out using as a framework the ethical domain of the HTA Core Model for the production of core HTAs¹ (version 3.0), an analytical methodology for the identification and discussion of ethical and deontological issues related to the use of health technologies produced by the European Network for Health Technology Assessment.

As for patients' safety, we will start analyzing all the actors in the process that will confront each other starting from the common project objectives and from a shared value considering the point of view of the patient, the community, the socio-cultural traditions in which the healthcare intervention is provided. We will also check that the AI algorithms ensure independence, quality, quantity, integrity and relevance of the data, guarantee a certain level of transparency to verify, explain, justify choices and decisions concerning the selection of data sources and the processes by which these data are processed and created by these systems. This will lead to specific clinical risk control models, assessing also the safety issues, such as the risk of serious harm or a high risk of milder harms, the benefit-harm-balance, possible adverse effects or poor tolerability.

1.2.2.5 Establish two-way and secure connections between physical and virtual worlds

The connections are enabled using a number of technologies, such as network communication, cloud computing and network security. Firstly, networking technologies send disease ongoing data to the cloud to power the DT. Secondly, cloud computing enables the DT to be developed, deployed and maintained completely in the cloud, so that it can be conveniently accessed by both physicians and patients from anywhere with Internet access. Lastly, since DT data concerns privacy-sensitive data, it is critical to guarantee the security of connections. We address these issues at a small scale since we implement and deploy on a local network; nevertheless, we will guarantee privacy and security using anonymization and encryption, ready to be deployed in other more complex scenarios.

A DT qualifier is the rate at which data is exchanged between the physical system and the virtual representation. This update is commonly described as being real-time [32] where changes in the physical state are updated in the virtual representation nearly instantly. While such a qualifier represents an ideal DT, it is not practical in cancer care where different data streams may be available at different frequencies. Hence, we design a semi-seamless update process occurring as the different datasets become available or are selected to occur at set intervals based on the use case and the frequency in which a decision will need to be made. Data security will be also granted making the AI models resilient to processing data in-the-wild. We investigate how to mitigate difficulties given by missing data and modalities in different directions. First, we mitigate the lack of training data and limited computing power using transfer learning from popular models trained on computer vision tasks. As an alternative solution we exploit GANs, and StyleGAN in particular, to get the real data manifold without labeled data, useful also for data augmentation in case of an imbalanced dataset. Learning with missing modalities is a key challenge and, except for a few contributions [33], [34] using traditional statistic-based imputation and dropout that we consider, we investigate GANs and transformers. As we are transferring data in a network, we also improve the resiliency of our AI models in adversarial scenarios by detecting attacks and recovering from them.

1.2.2.6 Data collection

In this project we will collect disease-related data from various sources to be processed by the LC DT. Using this multimodal data the DT will make available high-throughput methods to generate for virtual scanning and for treatment planning. We now describe the data we collect for each main goal, noting that there is an approved Ethical protocol (registration number 16/19 OSS ComEt CBM) and a submitted emendament that allow's us to gather the data.

Virtual scanning (objective 1): We will retrospectively collect chest and total body CT scans of patients diagnosed with NSCLC in stages III and IV from [13]. For these patients we also collect PET scans. For stage IV patients we also collect MRI scans from different body districts, which are performed when searching for metastasis. Additionally, we will also retrospectively collect CT, PET and MRI scans from non-NSCLC patients which are useful to pre-train the AI models [35].

Inclusion criteria for NSCLC patients are: confirmed diagnosis of stage III-IVNSCLC, availability of chest CT, PET or MRI scans, and no prior history of cancer treatment. We do not need paired data across different modalities since we leverage on AI models relaxing this constraint (see section 1.2.2.1). Exclusion criteria include non-diagnostic scans, and other major lung diseases that may interfere with the analysis. Inclusion

¹ <https://eunetha.eu/wp-content/uploads/2018/03/HTACoreModel3.0-1.pdf>

criteria for non-NSCLC patients include no history of cancer, availability of chest CT and/or MRI scans, and no major lung diseases that may interfere with the analysis. We also plan to include unpaired publicly available datasets from different modalities, such as the NSCLC-Radiomics-Genomics dataset for lung CT scans², FDG-PET-CT-lesions³ for PET scans, and The Visible Human Project for MRI⁴.

Virtual treatments (objective 2): We first gather data from CLARO project that includes >350 NSCLC patients at stages III and IV, for whom we already collected CT scans at the time of the diagnosis, PET scans clinical data, genomic data, EHRs, under the aforementioned Ethical Committee authorization. Such data has been updated at every follow-up, registering the OS, any toxicity (according to CTCAE v6.0), metastasis, progression-free survival. We will extend this collection including MRI data (brain, liver, backbone) for stage IV patients. For stage III patients we will also include CT scans collected during the radiation treatment, the follow-ups as well as their treatment plans. We also plan to include external publicly available datasets to enrich the cohort, such as the NSCLC-Radiomics-Genomics dataset that, although containing a less amount of multimodal data per patient, can be useful for external validation.

Sample size: For virtual scanning sample size impacts the training stage of the AI models. Although it is generally accepted that the more the data the better the model, it is known that CycleGAN can be successfully trained with hundreds of samples, even thanks to using data augmentation. The size of the CLARO dataset already accounts for several hundreds of cases, so it is reasonable to include > 300 scans. Nevertheless, the inclusion of publicly available datasets makes available other hundreds of scans. For virtual treatments, the size of the CLARO dataset already satisfies the minimum sample size computed for binary outcomes and for time-to-event outcomes, according to [36].

The consortium will manage project data according to FAIR principles [37], guaranteeing findability, accessibility, interoperability, and reuse of digital assets. The FAIR approach will be implemented in the project through the definition of a Data Management Plan in task T.3.5, delivered at month 2 (D1.1), and updated throughout the duration of the project. Intellectual Properties Rights are managed as part of the Dissemination and Exploitation work package, and discussed in section 2.2.1.

1.2.3 Existing relevant European and national proposals

Project	Relevance
CLARO: CoLIABorative multi-sources Radiopathomics approach for personalized Oncology in non-small cell lung cancer (2018-2022)	Supported by UCBM in the 2018 Strategic Internal call, here we demonstrated the efficacy of AI techniques in improving the care of LC. IDEA is a natural evolution from which we will build upon the collected data and by developing DT for more accurate and effective AI-based tools for LC care.
FAIR: Future AI research (2023-2026)	PNRR project. It is highly relevant to our project, as both share a common focus on the development of novel AI models. This means that there is a natural synergy, with the potential for collaboration and knowledge-sharing between the two. COSBI is involved in this project.
Platform for Precision Medicine: Artificial Intelligence and Integrated Clinical Diagnostics (2022-2024)	Supported by MISE, it aims to facilitate the acquisition of data for scientific research and innovation by fostering Federated Learning. By using the data of this initiative, we can acquire additional data that can be useful for validating IDEA results on external datasets. COSBI, RO, DI are involved in this project.
D3-4 HEALTH: Digital Driven Diagnostics, prognostics and therapeutics for Sustainable Health care (2023-2025)	PNRR project. It also develops DTs to improve healthcare, but it does not consider LC. It is limited to wearable technologies, sensors, and biomarkers without any use of imaging data. Furthermore, it does not address the problem of missing modalities, which our project aims to mitigate by employing synthetic generation to enhance resilience via the virtual scanner. Lastly, while D3HEALTH only mentions models for diagnosis, expected response to a therapeutic plan, and risk assessment, our project also implements XAI techniques for better understanding and interpretation of results.

We also carried out a research project search on CORDIS platform, which collects the results of the projects funded by the EU. While there are funded projects on DTs in the fields of Industry 4.0⁵, to the best of our knowledge no projects are now running under EU umbrella on DT for healthcare. Outside Europe, there is: a

² <https://wiki.cancerimagingarchive.net/display/Public/NSCLC-Radiomics-Genomics>

³ <https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=93258287>

⁴ https://www.nlm.nih.gov/research/visible/visible_human.html

⁵ Twin4Twin: <https://cordis.europa.eu/project/id/101079180> and AI-Twilight: <https://ai-twilight.eu/>

project on DT for uterine cancer supported by Epworth Medical Foundation, Australia; a project on airflow DT at Oklahoma State University [8], a project on optimal policy determination in sequential systemic and locoregional therapy of oropharyngeal squamous carcinomas supported by the National Institutes of Health and the National Science Foundation, US [19].

1.2.4 Interdisciplinary aspects

Interdisciplinary research integrates information, data, techniques, tools, perspectives, concepts, and/or theories from two or more disciplines or bodies of specialized knowledge. Its goal is to advance fundamental understanding or to solve problems whose solutions are beyond the scope of a single discipline or area of research practice. Interdisciplinarity is increasingly used to tackle complex scientific questions and address large societal challenges like those motivating IDEA. There is much literature that suggests that combining different academic disciplines and techniques in a single research project favors breakthrough research and innovation. By reflecting and promoting this vision, the Consortium comprises actors with diverse profiles: well-recognized computer scientists, with a solid background in image visualization, AI, data mining, radiologists and experts in radiation oncologists, and high-tech industry partners.

Medical experts on the targeted pathologies (RO, DI) drive the definition of the technical specifications needed for DT usage, and support the collection and processing of multimodal data, together with the industrial partner. COSBI and industrial partner CDI provide the consortium with the visualization and AI expertise necessary to build and feed the DT. The findings of WPs IDEA will be then evaluated by the medical and ethics experts (RO, DI, BH) to investigate and quantify the effectiveness and end-user satisfaction of this project's innovative technology components. Moreover, the industrial partners will bring in the consortium their management and development expertise to guarantee cohesion and to focus the project on its objectives. The ambitious goals of IDEA can only be completed through such an interdisciplinary approach.

1.2.5 Project governance structure

The **PI**, Paolo Soda from COSBI, ensures activities coordination, project administration, financial reporting, including quality control of all project deliverables and reports, punctual and proper completion of project tasks, financial management, technical and legal compliance of the consortium to the project objectives and EU regulations and efficient and effective communication among partners. The PI will lead a **Project Coordination Team** at COSBI that will coordinate and overview all administrative and financial obligations, supporting the planning, organization, and follow-up of project meetings. The PI will appoint and lead a **Quality Assurance Team** and define and distribute a Quality and Risk Assessment Plan for all project participants, encouraging and verifying the adoption of standards and controlled procedures. The PI and the Quality Assurance Team will perform quarterly Quality and Risk Reviews as part of the **Project Management Board** meetings and agree on any necessary mitigating action. The PMB is composed of partner leaders. This board will foster fast and efficient decision-making mechanisms and participative management. The PMB will meet every month during the project period. The consortium will establish an **Ethics Advisory Board** that will provide consultancy on ethical aspects of our activities, including data collection activities, needs and requirements design, personal data protection, and liaise with the local ethics committee.

1.2.6 Gender dimension

The gender dimension has been considered carefully in the design of the IDEA project. Although LC is not associated with a particular gender, its incidence is rapidly rising among women, who are diagnosed at an increasingly younger age compared with men [20]. One of the main reasons for this rise is increased tobacco smoking among women throughout Europe. However, many non-smokers also develop this disease, and taken together the etiological landscape of LC is changing. Other risk factors implicated in the differential LC development in non-smokers are genetic predisposition, tumor histology, and molecular profile. Furthermore, we found that sex is useful for AI algorithms to predict LC prognosis [13]. For this reason, in IDEA we will collect patients' sex as semantic information for AI algorithms. Furthermore, the consortium acknowledges gender equality as stated in the EU regulation 2021/695 of 18/4/2021 establishing Horizon Europe and supports the objectives of the framework program on: a) gender balance in research teams, b) gender balance in decision-making and c) integrating gender/sex analysis in R&I content. The consortium is committed to encouraging equal opportunities for careers among women and men.

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2. Impact

2.1 Pathways towards impact

The vision: *Imagine a future where personalized healthcare will provide every individual with the right type of care in the right way and at the right time. Imagine that every patient has a DT: a virtual representation of his/her health status.*

This plays a significant role when a patient diagnosed with LC is meeting with the care team (radiologists, radiation therapists, oncologists, etc.). Although there are many LC cases, it can be complicated to decide on each individual's treatment strategy. This is where his DT comes in. It starts by using the patient's medical data to create a patient-specific biophysical model, including a virtual version of his/her lungs that looks, behaves, and reacts just like the real one. Through a seamless flow of data and advanced analytics capabilities enabled by this proposal, the care team can connect the patient's DT, or to similar cases, and select the therapy with the best possible predicted outcome, both in the short and long term. The care team can carefully plan the procedure by simulating various treatment scenarios on the patient's DT, enabling them to select the one with the optimal computed result. The chosen ideal scenario forms the basis for real-time intervention guidance. During the procedure, all actions taken, as well as any unforeseen situations, are processed in real-time into the patient's DT, to redirect if necessary. As a result, the patient receives the treatment at the right time in the right way, with the best possible outcome.

This is one of many potential cases in which the DT adds value to the healthcare system: from personalized diagnosis and therapy selection to procedure planning and guidance, tailored to the patient's physical characteristics, medical history, current condition and future needs.

This narrative summarizes the impact of this project, which is one of the first attempts worldwide to develop DTs in healthcare. It focuses on cancer, a leading cause of death, accounting for nearly 10 million deaths in 2020, nearly one in six deaths in the world. Among cancer, here we direct our research towards NSCLC, which embraces more than 85% of new cases. Despite advances in patient treatment, 1-year survival of patients treated with chemoradiation and immunotherapy is 69%, while when they are treated only with chemotherapy the 1-year survival is 49% [38], also because of the side effects of such treatments. Nevertheless, as WHO reports [39], “many cancers have a high chance of cure if diagnosed early and treated appropriately”: this is the target of our project, where the development of DTs for LC will pave the way to select the right therapy for the patient by simulating different treatment scenarios.

The main impact of IDEA will be for the benefit of patients suffering from NSCLC. It will give them virtual copies of their diseases that permit the care team to design personalized treatments, driven by the potential of AI to mine heterogeneous data, which is collected during the treatment and as the disease evolves. The DT will improve the level of interaction between physicians and patients, helping to get more informed decisions. To this aim, the integration of XAI will offer insights into the reasons why an AI algorithm is providing a specific output to support the decision process. This will be demonstrated in the field validation. By improving the diagnosis, prognosis, and treatment of LC, the DT system will contribute to a **healthier society** by reducing disease burden, increasing life expectancy, and enhancing the overall well-being of individuals affected by LC, IDEA will thus create a new way of designing the treatments for people affected by NSCLC where, in the virtual world, the care team can investigate the impact of different therapies without any disadvantage for the patients.

Indeed, a key **societal impact** of IDEA is to enhance medical imaging and treatment planning using advanced AI approaches. This will not only facilitate the adoption of better treatment plans, but also promotes a more sustainable use of resources: indeed, image-to-image translation allows for reducing the ionization risk for patients (as in CT scans) as well as the power consumption, which is not negligible. This is why IDEA also has an **environmental impact**. For instance, the average power consumption for each MRI and CT examination is equal to 19.9kWh and 1.2kWh [40], respectively. Here, there is a risk that the DT methods provided by IDEA for imaging and treatment planning are not as “performance-enhanced” as real images and data, for example, in terms of image quality reconstruction, which may hinder them from being used as support tools in cancer care. Extensive testing, including visual, qualitative, and quantitative evaluation of the new methods will mitigate this risk.

IDEA will have a **societal and scientific impact** by raising awareness of the potential value of DT and medical data collection, beyond the scopes already considered in the community. This will demonstrate the

contribution that data such as images, laboratory tests, and clinical information make to scientific research in the field of cancer care and technology development.

IDEA will have a **scientific impact**, contributing to specific scientific advances in different fields. To build the virtual scanner, will deliver new collections of diverse imaging data captured by different scans (CT, MRI, PET), and collected in retrospective data inclusion as well as from publicly available datasets. Being collected from heterogeneous image scanners, these data will lead the way to new research on image-to-image translation. Furthermore, they are unpaired data, i.e., each observation in one dataset cannot be paired with an observation in another dataset with a different image modality: while this alleviates the burden of collecting coupled images, it better fits the nature of existing repositories. This will also lead to unprecedented opportunities, both for researchers and practitioners, to identify new methodologies and to improve existing approaches to learn the medical image translation task under this relaxed constraint and use them to develop new AI methodologies even beyond the duration of the project. Furthermore, the resultant hybrid datasets will support much larger virtual and mixed clinical trials to evaluate imaging devices and algorithms in terms of accuracy, robustness, uncertainty and generalizability. When focusing on radiation therapy planning, the DT integrates heterogeneous data representing the picture of the patient's health status during the therapy. They will give a unique, multi-perspective view of such intents, they will foster research on both DRL and multimodal DL, an area of AI that needs efforts to address several open questions on which modalities are useful, how, where and integrate the multimodal representations learned by the deep models, even in the case of sequential data. There may be regulatory and legal barriers related to data sharing and privacy that may hinder the training of AI models employed in the DTs. Compliance with GDPR and other relevant regulations will be crucial to ensure the protection of patient data and privacy.

Furthermore, the need to explain to medical experts as well as to patients the reasons behind the information provided by the DT will push forward research in the field of XAI, an area where there has been growing interest in recent years but where these efforts have only been directed towards unimodal learning, whilst this project deals with multimodal data. This has also the immediate purpose of providing socially beneficial and responsible AI, complying with the GDPR “Right to explanations”. Hence, such technological solutions will impact the deployment of the released models by other researchers in terms of usability, control, inspection, and quantifiable robustness. Moreover, employing XAI would allow the active involvement of expert physicians in the interpretation of the models' decisions, including the possibilities of extracting new knowledge about links between the models' inputs and outputs that a human observer alone can't derive. This novel and controlled way of modeling and understanding human behavior have the potential to inform up-skilling training programs for medical experts, giving new inputs on the LC disease considered in this study.

In this respect, the involvement of medical experts in the interpretation of XAI explanations would serve as mitigation for possible mismatches between the actual explanation complexity and fairness together with the user's understanding and opinion.

As for data reusability in the future, one possible obstacle is the lack of organization, pre-processing, and documentation of the collected material. Avoiding this risk requires the establishment of systematic data management protocols that recognize, from the project's onset, the exquisite scientific and societal value of the collected data. Another possible obstacle to the future exploitation of the collected data is related to their usefulness to nurture future technology development. Some data formats and measurement granularities may become obsolete and of scarce usefulness compared to future recording and storage techniques. However, considering the type of data that will be collected, this is a very remote risk. Proposing and supporting the use of the collected data as a standard for the target clinical communities (well-represented in the consortium) will help to mitigate it.

IDEA will deliver a new **affordable method to create a medical DT**, using new AI methodologies but also off-the-shelf devices from the consumer market. This will have **an impact that is both technological and economic**. Moving from our experience on LC, **our methodology can be easily extended and tested on other tumors**, so that affordable DTs may become more desirable to hospitals and cancer care teams to build patient realistic twins to simulate different image scans and possible therapeutic plans. Indeed, DTs can also impact the organization and sustainability of healthcare institutions. Potentially, new (small-sized) companies will be able to offer services for the creation of DTs - without the need of making a significant investment but exploiting the results of this project. A possible obstacle here is the rapid development of AI methodologies available, but this issue should be considered an opportunity rather than a limitation. In fact, since the partners of this project are AI experts, they will be able to exploit any innovation in this field to improve the decision

support algorithms offered in the DTs. Even before the end of the project, algorithms selected by the partners and used here may decline in their market acceptance and availability. It is necessary to build a community around the software solution that can keep the software up-to-date and compatible with new devices as they reach the market in future years.

2.1.1 Impact for UCBM

IDEA develops DTs of patients, a digital replica emphasizing the uniqueness and centrality of human beings that will permit to design personalized treatments to get better outcomes.

This project fits well the goals of UCBM's strategic plan, including strengthening the areas of research focusing on biomarkers, hospital 4.0, and the circular economy. Turning the attention to UCBM professionals, IDEA will facilitate cooperation between different areas. Moreover, by improving UCBM's innovation potential, IDEA will provide researchers and students with opportunities to develop new knowledge and expertise in AI, medical imaging, and DTs. This will lead to the development of a highly skilled workforce that is well-equipped to meet the challenges of the rapidly evolving healthcare industry. There are also other three main benefits for UCBM: i) the value of the project is beyond the requested grants, thanks to the participation of enterprises that will make available in-kind resources, ii) the topic addressed in this project perfectly fits Horizon Europe pillars (Table 1.2) so that we can expect to enlarge the Consortium, fit the goals of the project to a wider extent and apply to forthcoming EU calls, iii) IDEA is the natural continuation of the previous project CLARO supported by the Internal call in 2018, where we investigated radiopathomics for personalized oncology in NSCLC, and IDEA leverages on the results attained in such a project.

2.1.2 Scalability

The results of the project will impact the LC therapeutics market, which globally accounts for USD 29.8 billion in 2022 and it is projected to hit around USD 55.6 billion by 2030 with a CAGR of 8.11% between 2022 to 2030 [39]. Furthermore, the healthcare DT market in 2022 accounts for USD 572 million, and it is projected to reach USD 3.5 billion by 2030 at a CAGR of 25.6% during such a period [39]. Hence, the development of IDEA fits this period with impressive, expected growth. Since the development of the first DT, methodological work on empowering this technology with AI and on the use to deploy personalized medicine in LC care has been relatively sparse. However, recent advancements have provided new evidence that leads to considering these technologies as ever more promising to find the right therapy for the right patient. The results of IDEA are expected to address issues related to achieving realistic solutions which will be scalable for new-generation oncology.

2.2 Measures to maximize impact - dissemination, exploitation and communication

2.2.1 Dissemination and communication plan

Dissemination and communication activities will be carried out throughout IDEA. **Follow-up plans** will be developed addressing the innovative outcomes of IDEA. In addition to traditional approaches, we will make extensive use of digital communication means to provide up-to-date information. Table 2.1 shows the dissemination and communication plan related to different target groups. It focuses on explaining to all the population levels the usefulness of this technology, the benefit for people and society and thus the importance of DTs development in healthcare.

Table 2.1. Dissemination and communication strategy for different target groups

Target group	Channel	Content	Strategy
- Consortium members - Scientific Community - General Public - Industry	Website (continuously during the project)	- Initial contact point with the project (objectives, workplan, consortium) - go-to place for consortium - DT use cases information/updates.	- Visualization (videos, webinars) - Descriptive languages (press material)
	Newsletter (every 6 months)	Short updates and longer communications on the project	Descriptive languages
- General Public - Physicians - Patients - Industry	Social media (Linkedin, Twitter, every month)	General messages to communicate with external researchers and the general audience	Storytelling, Visualization
Scientific community	Arxiv, Google Scholar, ResearchGate, Partner Search EU Portal	Share papers, ask/answer questions and search for collaborators	Publications, talks
Business oriented	LinkedIn (every month)	Professional networks, linking experts	Scientific results for

profiles		from different areas	large audience
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The *scientific dissemination* will be maximized by publishing the project's results, whenever possible, in open access journals or opting for granting open access rights. Table 2.2 lists relevant targeted conferences and journals. We plan to use a mix of “gold” and “green” models to publish the results of our research. Knowledge will be managed and protected by means of dissemination rules agreed by the Consortium. Only after a validation of these dissemination rules, the knowledge generated by the project will be allowed to be disseminated and shared with different interest groups, depending on the scope of the information.

Table 2.2. Targeted Peer-reviewed international conferences and journals to disseminate the scientific results.

Topic	Conferences	Journals
DT system	IEEE Int Symp on Computer Medical Systems, IEEE Int Conf on Engineering in Medicine and Biology, IEEE Int Conf on Bioinformatics and Biomedicine	Nature Machine Intelligence, IEEE Access, Decision Support Systems, IEEE Internet Computing
AI algorithms	Int Conference on Pattern Recognition, MICCAI, Int. Conference on Image Analysis and Processing, IEEE Int. Symp on Biomedical Imaging	Artificial Intelligence in Medicine, Pattern Recognition, IEEE Transaction on Medical Imaging, Medical Image Analysis, Information Fusion, Expert Systems with Applications
Cancer disease	International Association for the Study of Lung Cancer, European Society of Radiation Oncology, Associazione Italiana di Radioterapia e Oncologia Clinica, Società Italiana di Radiologia Medica, European Society of Medical Oncology	Plos One, Nature Medicine, Scientific Reports, Radiology, Radiotherapy Oncology, IJROBP, Lancet Digital Health

Table 2.3 summarizes all outreach activities in line with IDEA’s project rationale, mission, principles, and goals by employing channel selection coupled with targeted messages and content, and it includes the organization of workshops to disseminate the results.

Table 2.3. Summary of dissemination and communication activities

Stakeholders and participants: - Consistent analysis of stakeholders and participants will be carried out throughout the project; - Proactive outreach via prompt communication with stakeholders and participants, also via regularly personalized emails; - Workshop organization: at least one at the end to disseminate the project results.	Design and branding: - Coherent branding will be used (logo, infographics, templates, etc.); - Diversified communication materials (gadgets, brochures, posters, etc.) will be developed; - UCBM funding acknowledgment will be used.
Messages and contents: - Definition and formulation of key messages, languages, contents, guidelines, and directions for the different audiences. Examples of key messages are: Personal data rights are respected, research leads to useful DTs, reduced costs to individuals & society; - Consortium know-how will generate high-quality dissemination material and paper throughout the project, with regularly updated and targeted content provided to each stakeholder group.	Multimedia and educational: - Social networks (Twitter, LinkedIn, ResearchGate) multimedia communities (Youtube) and a publication repository will be established and populated with relevant multimedia; - Project results will be presented to public general audiences in fairs for a positive impact; - Thematic educational events will be organized for young professionals in oncology and computer scientists to engage more researchers in the project domains.
Channels and publications: - Different channels will be used to share research results and scientific messages through appropriate journal selection and dissemination through events and activities (conferences, workshops, etc); - Major publications, also in open access, will be accompanied by dedicated press releases to science news agencies and portals to promote media coverage; - In addition to the website and social channels, results and output will be shared via existing partners outlets to extend access to stakeholders; - An agreement with the Consortium will be sought to	Online and paper material: - Website and social media will be regularly updated with project news and achievements; - Accessible publications specifically designed for non-technical audiences and patients will be produced; - The project newsletter, distributed via a subscribed mailing list, will report project achievements to a broad audience with a common language, large use of images and infographics; - Promotion of different results will also be conducted via broadly informative publications.

share reciprocal channels to disseminate and communicate results and updates.	
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The *evaluation of the communication and dissemination activities* will be performed qualitatively and quantitatively.

The success of the communication and dissemination activities will be determined by the potential outreach and impact. This will include for example a) the number of dissemination events for each target group (e.g. patients, healthcare professionals, policy-makers and industry); b) the number of participants at each event; c) questionnaires filled by the participants to address the impact of the event on the specific target group; d) visibility, including contact with public media and exposure to a large audience.

A specific evaluation will address potential exploitation activities (e.g. registrations, spin-offs or solutions reaching the market) generated by the dissemination of the results of the project. The communication and dissemination activities will last for the overall duration of the project and beyond. After the end of the project, the web page will be kept active, as well as the social network profiles, and active communication actions will be delivered for at least one extra year.

2.2.1 Exploitation planning and management of intellectual property rights

IDEA has a specific work package task (T5.2) dedicated to the exploitation of project outputs, where we analyze the collected evidence and user feedback of participating physicians to assess the implications of the tools for applications improvements, as well as their extendibility to other tumors. The work carried out in this task will include a full analysis of both the research and market landscapes, including ethical and regulatory aspects. To enable wider exploitation, we will prepare a report documenting how the resulting tools can be deployed and fully implemented by external users other than the IDEA team. Finally, an innovation forum will be set up to enable future innovations and collaborations through the IDEA stakeholders, and contiguous project calls, and generate effective value propositions for European technologists, businesses, and industries. Furthermore, with reference to the *exploitation of the research data* collected during the project, their preservation will be handled in accordance with national legislation. The data collected by this project will be made available to researchers in anonymous form upon presentation of a formal study proposal that will be evaluated by the Partners' Board.

Intellectual Property Rights Management

The rules and regulations concerning the management of Intellectual Property Rights will be set forward in a Consortium Agreement to be signed among project partners before the start of the project. The main purpose of the Consortium Agreement is to establish a legal framework for the project in order to minimize conflicts. In order to ensure a smooth execution of the project, the following principles will be established:

- The project partners shall grant royalty-free access to Background and Foreground IP, if it is technically essential for carrying out the tasks of the project. Therefore, Background IP which is technically essential for carrying out the project as well as Foreground IP which is explicitly excluded from the project will be identified in the Consortium Agreement;
- Any intellectual property rights in any Background and/or Foreground IP made available by a project partner in the course of the project shall remain of exclusive ownership of that project partner.

Any details on access rights to Background and Foreground IP will be defined in the Consortium Agreement.

3. Implementation

3.1 Work plan – Work packages, deliverables and milestones

The project is structured according to several main phases partially carried out in overlapping periods, as described in the following, with the objective to:

1. Set up
 - a. Start the Consortium with the definition of the objectives, commitments, responsibilities, composition of the management bodies, based on the information presented in this proposal;
 - b. Plan the project and the activities, define, identify, assign or reserve the resources for the different teams;
 - c. Enforce team spirit, leveraging on previous well-established cooperation among the RUs;
2. Research, design and development
3. Experimentation
 - a. To quantitatively assess the performance of the DT and of its AI algorithms;
 - b. To analyze the impact (ethical, economical, etc) of this technology in healthcare;
4. Assessment and evaluation
 - a. To verify whether and to what degree the project has achieved the expected impacts;
5. Shut-down
 - a. To ensure the greatest possible scientific, business, economic, and social impact of the results;
 - b. To evaluate possible subsequent activities or projects.

The activities will be carried out at the Università Campus Bio-Medico di Roma. Furthermore, the external institution CDI SpA will cooperate in the project development: it will be involved in the needs definitions, data collection and in the evaluation. For this reason, and to make clearer the description of project implementation and partner relationships, in the WPs and tasks we also include CDI, with an estimate of its monthly efforts offered in terms of in-kind support.

The project duration is expected to be 24 months, and below we report: (i) the detailed workplan, with the work packages (WPs) and the details on the composing tasks (Ts); (ii) A Gantt diagram showing the timing of the different work-packages and their main tasks (Figure 4); (iii) the PERT diagram of the interrelations between the WPs (Figure 5); (iv) the milestones; (v) the risk analysis.

Figure 3 offers a schematic view of the work packages and of the composing tasks.

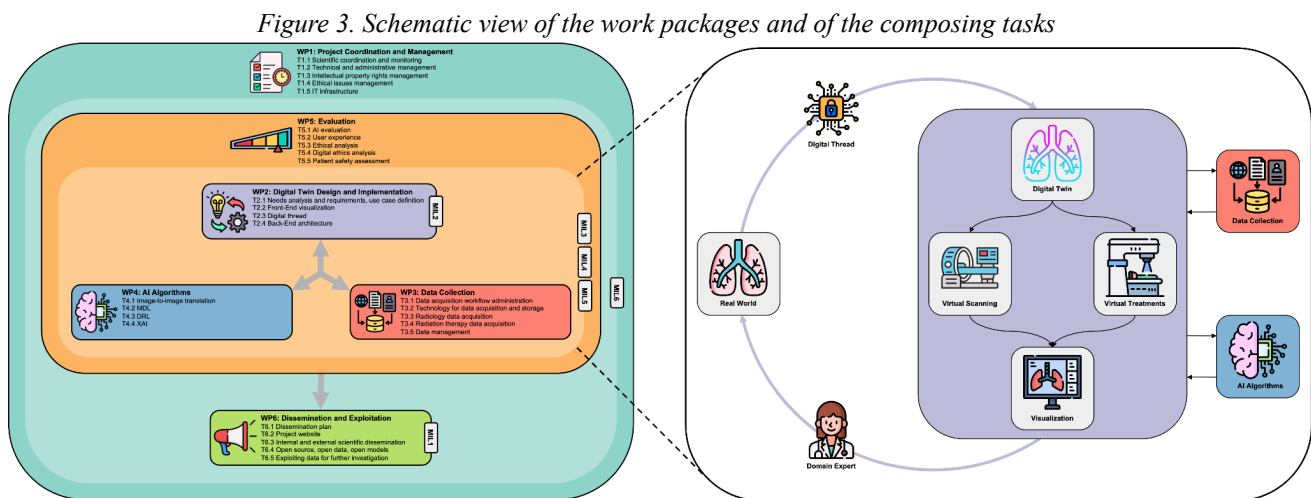


Figure 4. Gantt Chart

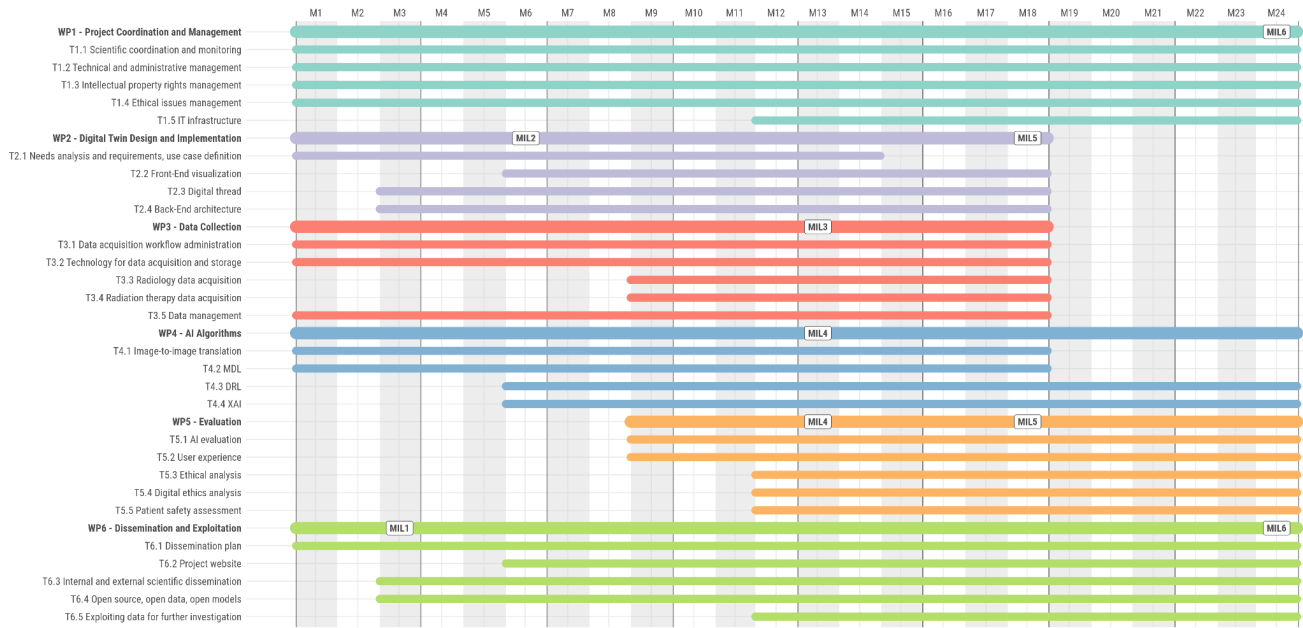


Figure 5. PERT Diagram

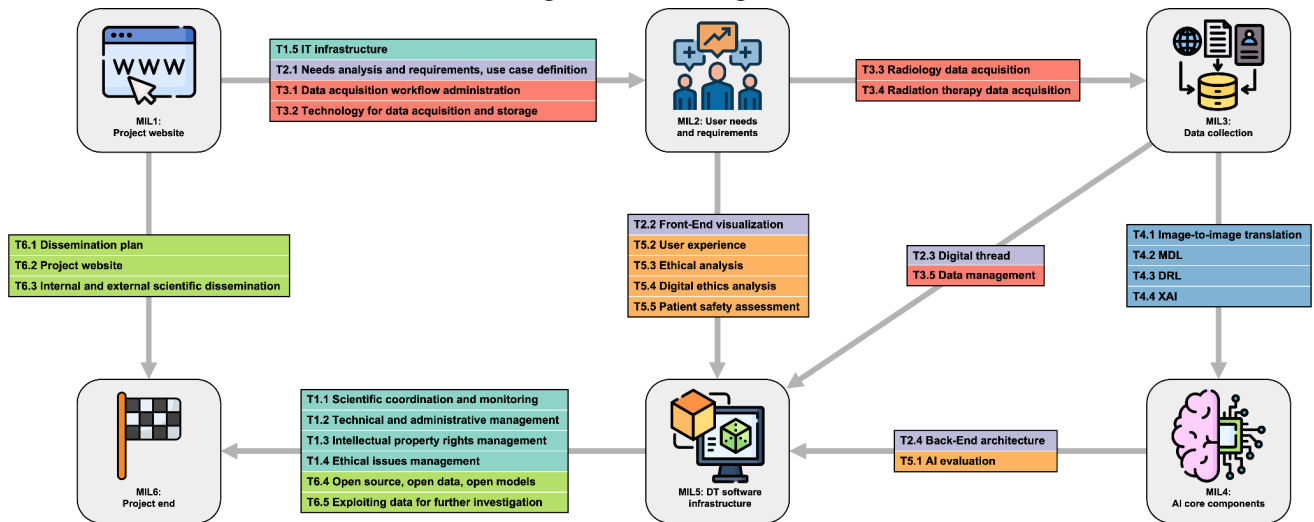


Table 3.1a: Work Package Description

Work package number	WP1	Start date or starting event				1
Work package title	Project Coordination and Management					
RU number	1	2	3	4	5	
RU name	COSBI	RO	DI	BH	CDI	
Person/months per participant	2	1	1	2	2	
Objectives Ensure effective project planning, coordination, and management, as well as to monitor the project's progress and outcomes. This work package is critical for the success of the project, as it helps to ensure that the project is completed on time, within budget, and meets the project's objectives. Furthermore, this WP will also develop the IT infrastructure.						
Description of work The WP is under COSBI responsibility. All the partners will participate, with adequate commitment, in carrying out tasks and activities that constitute the WP. The tasks of the WP are (highlighted the lead partner for each task): T1.1 Scientific coordination and monitoring (COSBI, RO, DI, CDI)						

This task involves managing the scientific aspects of the project, ensuring that the research and development activities are carried out according to the project plan and objectives. It includes coordinating the project's scientific activities, monitoring the progress of the research and development, and ensuring that the project outcomes align with the scientific objectives. Additionally, the task involves organizing project meetings, developing scientific reports, and communicating the project's scientific progress to stakeholders.

T1.2 Technical and administrative management (COSBI, RO, DI, CDI)

This task involves the overall management of the project, including administrative and technical aspects. It includes setting up the project team, defining roles and responsibilities, establishing communication protocols, and ensuring that the project is executed within the defined timeline and budget. Additionally, it involves overseeing project finances, managing project risks, and ensuring that the project meets quality standards.

T1.3 Intellectual property rights management (COSBI, RO, DI, CDI)

Intellectual Property Rights management is crucial for protecting the project's intellectual property and ensuring that the project outcomes are adequately protected. The task involves identifying and managing the project's intellectual property, including patentable inventions, software, and other intangible assets.

T1.4 Ethical issues management (COSBI, RO, DI, BH)

Here we manage ethical issues related to the work carried out, dealing especially with the issues related to the use of patients' multimodal medical data. This task includes developing ethical guidelines for the project, obtaining ethical approvals from relevant authorities, and ensuring that the project's activities are conducted in an ethical and responsible manner. Security measures for storage and handling users' personal data will be adopted.

T1.5 IT infrastructure (COSBI, RO, DI, CDI)

IT infrastructure is essential for the successful implementation of the project. This task involves setting up and managing the IT infrastructure required for the project, including hardware, software, and network systems. Additionally, the task involves ensuring that the project's IT infrastructure is secure, reliable and meets the project's requirements. It includes managing data storage, data security, and data privacy issues, and ensuring that the project's IT infrastructure is compliant with relevant regulations and standards.

Deliverable no.	Deliverable title	Delivery date
D1.1	Plan and procedures	M2
D1.2	Annual meeting year 1	M13
D1.3	Closing session documents and follow-up planning	M24

Work package number	WP2	Start date or starting event				1
Work package title	DT Design and Implementation					
RU number	1	2	3	4	5	
RU name	COSBI	RO	DI	BH	CDI	
Person/months per participant	11	2	2	1	2	

Objectives

WP2 aligns the technical innovations to be delivered by IDEA to the needs and preferences of the target users. Given the project's overall goal of developing DTs for NSCLC boosted by AI tools, this WP initially explores existing solutions and workarounds relevant to this goal. Through interaction with technology development tasks, it represents the interests of end-users as primary stakeholders for the technology. It undertakes the co-design of requirements and specifications for usability and interaction of the technological developments. In this RO, DI and CDI are referred to as clinical partners, collecting input and feedback on the DTs.

Description of work

The WP is under RO's responsibility. The tasks of the WP are (highlighted the lead partner for each task):

T2.1 Needs analysis and requirements, use case definition (COSBI, RO, DI, CDI)

Partners investigate the requirements and the use cases for the DT that guide the development process (RO, DI). This involves identifying specific scenarios in which the DT can be used to simulate the NSCLC behavior and response to different treatments (RO, DI), as well as the graphical interfaces (COSBI).

T2.2 Front-End Visualization (COSBI, RO, DI, CDI)

The development of a visualization interface for the DT involves designing a user-friendly interface that allows healthcare professionals to interact with the DT and visualize the simulated behavior and response of the LC to

different treatments. The interface will need to be intuitive, easy to use, and provide the necessary information to make informed treatment decisions.

T2.3 Digital thread (COSBI, RO, DI)

COSBI establishes a bidirectional data exchange between the DT and the patient's medical records. This will enable the DT to access relevant patient data, such as medical history, symptoms, and diagnostic results, in order to simulate the behavior and response of LC to different treatments. It will also allow the DT to update the patient's medical records with the results of the simulations. RO and DI support the assessment of the technology developed here.

T2.4 Back-End architecture (COSBI)

This task develops a cloud-based medical imaging and simulation platform for the DT. The platform provides access to: advanced algorithms that can generate models of the lung from the medical images; web-based medical imaging services, such as CT, MRI and PET, which can be accessed remotely by healthcare professionals to simulate the scanning of the lungs; web-based simulation services, where different virtual treatments, such as chemotherapy, radiation therapy, or surgery, can be simulated on the models of the lung generated from the medical images.

Deliverable no.	Deliverable title	Delivery date
D2.1	Needs, requirements, and use cases	M6
D2.2	Development of a DT software framework	M14
D2.3	DT user manual	M24

Work package number	WP3	Start date or starting event				1
Work package title	Data Collection					
RU number	1	2	3	4	5	
RU name	COSBI	RO	DI	BH	CDI	
Person/months per participant	5	7	6	3	1	

Objectives

Central to this project is the data used for training AI-based algorithms. Section 1.2.2.6 details several aspects of data collection. Essentially, multimodal data (cf. table 1.3 in section 1.1.4) are acquired from included patients. This WP collects the data and prepares it for consumption by machine and DL algorithms in WP4. Data collection continues over most of the lifetime of the project, reflected in the duration of this WP's tasks. Data collected in this WP is released according to what is stated in T6.4.

Description of work

The WP is under RO's responsibility. The tasks of the WP are (highlighted the lead partner for each task):

T3.1 Data acquisition workflow administration (COSBI, RO, DI, BH)

RO and DI design and COSBI implements a data acquisition workflow that ensures the collection of high-quality and relevant data. This will involve identifying the sources of data, and establishing protocols for data acquisition. RO and DI interact also with the Ethical Committee to ensure that all data collection procedures are in line with ethical standards and protect the privacy and confidentiality of the patients involved. In addition, COSBI will retrieve public data from various repositories. It is important to ensure a standardization procedure to reduce the variability in the quality and consistency of the data obtained from different centers.

T3.2 Technology for data acquisition and storage (COSBI, RO, DI, CDI)

COSBI, RO and DI select the appropriate technology for data acquisition. This could include various types of medical imaging equipment, such as CT, PET and MRI scanners; COSBI selects the software and hardware for data capture and storage. It is important to choose the most appropriate technology that is capable of generating high-quality data and can seamlessly integrate with the AI algorithms.

T3.3 Radiology data acquisition (COSBI, DI, CDI)

DI acquires and organizes radiology data, that will be used to train the AI models, as described in section 1.2.2.6 "Data collection". The system for collecting and storing the data is defined in T3.2. DI ensures that the data is of high quality and annotated correctly. DI efforts will be directed toward obtaining a large and diverse set of radiology data to ensure that the AI models are effective in different scenarios. COSBI offers technological support.

T3.4 Radiation therapy data acquisition (COSBI, RO, CDI)

RO acquires and organizes radiation therapy data, such as treatment plans, dosimetry records, periodic CT scans, which will be used to train the AI models (section 1.2.2.6 "Data collection"). COSBI offers technological support. RO

efforts will be directed towards obtaining large and diverse set of radiation therapy data to ensure that the AI models are effective in different scenarios.

T3.5 Data management (COSBI, RO, DI)

This task involves managing the collected data and ensuring its quality and integrity. COSBI and BH develop a data management plan that includes data storage, retrieval, and analysis procedures. It is important to establish protocols for data backup and security to ensure that the data is protected from loss or theft. Additionally, it is crucial to ensure that the data is easily accessible and can be shared among researchers and clinicians involved in the project.

Deliverable no.	Deliverable title	Delivery date
D3.1	Data management and workflow administration	M2
D3.2	Report on data collection	M13

Work package number	WP4	Start date or starting event				1
Work package title	AI Algorithms					
RU number	1	2	3	4	5	
RU name	COSBI	RO	DI	BH	CDI	
Person/months per participant	21	2	1	1	0	

Objectives

This WP provides machine learning- and DL-based algorithms as software building blocks to enrich the DT to predict the NSCLC behavior and disease outcomes. Work in this WP requires expertise in statistical model design, DL, MDL, and XAI. It acts as a bridge between WP2 (DT design and implementation) and WP3 (data collection). Its goal is to build an AI-powered DT to provide enhanced diagnosis and treatment decision support, as well as a more individualized and patient-centered focus. The models are evaluated within the tasks of this WP, but also as part of an integrated system in WP5. The source code written in this WP is released according to the plans stated in T6.4.

Description of work

The WP is under COSBI's responsibility. The tasks of the WP are (highlighted the lead partner for each task):

T4.1 Image-to-image translation (COSBI, DI)

Here COSBI will develop image-to-image translation algorithms. These algorithms will be used to convert medical images of the lung, from one modality to another, that can be used by the DT. Section 1.2 already presented the methodology: shortly, we will use generative DL techniques to learn the mapping between the different image modalities and generate high-quality images that can be used to simulate the behavior and response of the LC to different treatments (Pix-2-Pix, CycleGAN). Pre-training on non-NSCLC patients will be useful for a faster training phase. Evaluation includes qualitative and quantitative evaluation: the former relies on the Visual Turing test; the latter relies on computing paired and unpaired metrics. Standard hold-out, cross-validation and bootstrap are the statistical means to compute the performance.

T4.2 MDL (COSBI)

COSBI develops MDL algorithms, which will be used to combine different types of data, such as medical images, EHRs, clinical data, that aim to predict treatment outcomes, i.e., the OS. The algorithms will be designed to be resilient to data in-the-wild, such as missing data and noise, as described in the methodology (section 1.2). They will use advanced DL techniques, such as attention mechanisms, to model complex interactions between different data modalities. Evaluation is based on computing Ct-index usings hold-out, cross-validation and bootstrap as the statistical means to compute the performance, and RO and DI will assess the performance achieved.

T4.3 DRL (COSBI, RO)

COSBI develops DRL algorithms, which will be used to optimize treatment decisions by learning from the simulated behavior and response of the LC to different treatments in the DT. The algorithms will use a trial-and-error approach to find the best treatment strategy, taking into account the patient's individual characteristics and medical history. Evaluation is carried out by COSBI and RO, and it directly compares the policies proposed by the DRL approach and the standard radiation therapy fractionation scheme, testing the different treatment policies on a random population of patients generated by the virtual radiotherapy environment.

T4.4 XAI (COSBI, RO, DI)

Here COSBI develops XAI algorithms to offer fair and trustworthy information to the DT, boosting the clinical adoption of AI in healthcare. To provide healthcare professionals with a transparent and interpretable view of the AI decision-making process, here we implement the methodologies described in section 1.2. They will be used to explain the treatment recommendations made by the DT, and to provide insight into the features and data that were used to

make these recommendations. In cooperation with RO and DI, evaluation is performed considering quantitative and qualitative metrics, such as stability of the explanations, number of rules, number of features, visual Turing test, etc.

Deliverable no.	Deliverable title	Delivery date
D4.1	Report on core AI components and performance	M12
D4.2	Report on core AI components and performance	M24

Work package number	WP5	Start date or starting event				9
Work package title	Evaluation					
RU number	1	2	3	4	5	
RU name	COSBI	RO	DI	BH	CDI	
Person/months per participant	8	2	2	1	2	

Objectives

WP5 will develop methods and undertake studies to evaluate the success of IDEA's innovative technology developments and will explore precisely what the term "success" means in this context. In this WP, RO, DI and CDI plan and conduct evaluation studies with the clinical context. COSBI evaluates the performance, effectiveness, and impact of the LC DT combined with AI developed in the previous WPs, and summarizes the continuous evaluation already included in WP4. This WP is critical to ensure that the proposed solution is safe, effective, and ethical, and that it meets the needs and expectations of patients and healthcare professionals.

Description of work

The WP is under BH's responsibility. The tasks of the WP are (highlighted the lead partner for each task):

T5.1 AI evaluation (COSBI, RO, DI, CDI)

COSBI evaluates the performance of the AI algorithms during the development cycle, as described in WP4. Furthermore, here COSBI carries out the final evaluation even considering external validation datasets of medical images and clinical data, mentioned in section 1.2.2.6 and in WP3. Performances scores depend on the task, such as signal-to-noise ratio and Peak signal-to-noise ratio for virtual scanning, Ct-index for OS survival prediction, the reward for DRL. The evaluation will also include benchmarking the AI algorithms against state-of-the-art approaches and identifying areas for improvement. Quantitative scores will be assessed by RO, DI and CDI, and they are also involved in the qualitative evaluation.

T5.2 User experience (COSBI, RO, DI, BH, CDI)

This task is to evaluate the user experience of the LC DT: BH will conduct user studies and surveys to gather feedback from users on the usability, usefulness, and satisfaction of the proposed solution, thus involving RO, DI and CDI. The evaluation will also include identifying barriers and facilitators to the adoption and implementation of our solution.

T5.3 Ethical analysis (COSBI, RO, DI, BH)

BH develops an ethical and moral analysis of the technology's use and impact on the physicians and patients, evaluating the ethical implications of using the LC DT combined with AI. Categories of medical ethics will be applied in modeling the space of choices and freedom allowed by the DTs. This study will be confronted with a thorough analysis of the user's experience and of the professionals' work and accountability. This task involves assessing the ethical, legal, and social implications of using DT in healthcare, such as privacy, confidentiality, bias, and accountability. The evaluation will also include developing guidelines and best practices to ensure that the proposed solution is aligned with ethical principles and regulatory requirements. The other participants will contribute to this task with reference to their skills and roles in the project. Ethical and deontological framing will be carried out using as a framework the ethical domain of the HTA Core Model⁶ for the production of core HTAs (version 3.0), an analytical methodology for the identification and discussion of ethical and deontological issues related to the use of health technologies produced by the European Network for Health Technology Assessment (EUnetHTA).

T5.4 Digital ethics analysis (RO, DI, BH)

BH will compare with national and international privacy standards the project's data protection and security, property, circulation and storage procedures. The reflection will reach deeper since digital ethics is the field of study concerned with the way technology is shaping our political, social, and moral existence. The task will address and quantify the risks and possible unintended consequences. RO and DI will participate in this task.

T5.5 Patient safety assessment (BH, RO, DI)

⁶ <https://eunethta.eu/wp-content/uploads/2018/03/HTACoreModel3.0-1.pdf>

The new technology based on AI will have to guarantee independence, quality, quantity, integrity and relevance of data as well as a certain level of transparency to verify, explain, justify choices and decisions regarding the selection of data sources and the processes through which these data is processed. Furthermore, the objective of patient safety must be pursued through the contribution of all those involved in the construction of the new technology and in the revision of the care process in which it will be introduced. Patient safety framing will be carried out using as a framework the safety domain of the HTA Core Model for the production of core HTAs (version 3.0).

Deliverable no.	Deliverable title	Delivery date
D5.1	Report on AI evaluation and DT user experience	M24
D5.2	Report on the ethical analysis	M20
D5.3	Report on patient's safety	M22

Work package number	WP6	Start date or starting event				1
Work package title	Dissemination and Exploitation					
RU number	1	2	3	4	5	
RU name	COSBI	RO	DI	BH	CDI	
Person/months per participant	9	2	2	2	2	

Objectives

Ensuring that the results and outcomes of the LC DT are effectively disseminated and exploited. This work package aims to maximize the impact of the project by sharing the knowledge, resources, and tools developed in the previous work packages with relevant stakeholders, such as healthcare professionals, researchers, patients, and industry.

Description of work

The WP is under COSBI responsibility. The tasks of the WP are (highlighted the lead partner for each task):

T6.1 Dissemination plan (COSBI, RO, DI)

This task is to develop a dissemination plan that outlines the objectives, target audiences, channels, and timelines for disseminating the project results and outcomes. This plan will be developed based on the project objectives and will identify different groups that will benefit from IDEA's results, defining the best communication tool for each one. The dissemination plan will also include a communication strategy that identifies the key messages, branding, and storytelling elements to effectively communicate the project's value proposition and impact.

T6.2 Project website (COSBI, RO, DI)

COSBI will set up and maintain a dedicated domain where individual consortium participants will be able to add/edit content. This will ensure that all material on the site will be actively updated. It will serve as a central hub for disseminating project information, updates, and resources. There will be a public area, to disseminate to the large public the project's objectives, partners, methods, results, and outcome. The website will also provide access to project reports, publications, datasets, and software tools developed. There will be also private members' area where research material will be exchanged among partners. The management and updating of the website will last for the whole duration of the project. All partners will diligently contribute to keeping the public area updated.

T6.3 Internal and external scientific dissemination (COSBI, RO, DI, BH)

This task is to disseminate the project's scientific findings and outcomes through various channels, such as scientific publications, conference presentations, workshops, and webinars. This task involves identifying relevant scientific venues and audiences, preparing manuscripts, abstracts, and presentations, and engaging with scientific communities and networks. The dissemination of scientific outcomes will enhance the project's visibility, credibility, and impact, and facilitate the uptake and replication of the project's methods and tools. All partners are involved.

T6.4 Open source, open data, open models (COSBI, RO, DI)

All data collected in WP3, the source code written for the project in WP2 and WP4, and the AI models developed in WP4 will be released under Open Source licenses, like MIT, Apache 2.0 or CC-BY 4.0. Possible open repositories and platforms are GitHub, Zenodo, and Figshare. The stable versions of data and code, which will be at least 80% of the total, will be made available by the end of the project; the rest, if any, within the two successive years. The data will be fully anonymized to ensure privacy protection.

The dissemination of open resources will facilitate the replication, reuse, and improvement of the project's methods and tools, and foster collaborations and innovation in the field. All the partners are involved.

T6.5 Exploiting data for further investigation (COSBI, RO, DI, BH, CDI).

Each content and data collected in the project will be analyzed by all partners to discover how to extend the project beyond its life, and to discuss possible next applications to support the research and the exploitation.

Deliverable no.	Deliverable title	Delivery date
D6.1	Dissemination Plan and Website creation	M3
D6.2	Efficacy monitoring of communication to the general public via project website and social networks	M14
D6.3	Efficacy monitoring of communication to the general public via project website and social networks	M24

Table 3.1b: List of work packages (colors refer to different WP according to Figure 3)

Work Package No	Work Package title	Lead RU No	Lead RU name	Person months	Start month	End month
1	Project Coordination and Management	P1	COSBI	8	1	24
2	DT Design and Implementation	P1	COSBI	18	1	14
3	Data Collection	P2	RO	22	1	18
4	AI Algorithms	P2	COSBI	25	1	24
5	Evaluation	P3	DI	15	9	24
6	Dissemination and Exploitation	P1	COSBI	17	1	24
				105		

Table 3.1c: List of deliverables (colors refer to different WP according to Figure 3)

Deliverable (number)	Deliverable name	WP number	Lead RU name	Type	Dissemination level	Delivery date
D1.1	Plan and procedures	WP1	COSBI	R	CO	M2
D3.1	Data management and workflow administration	WP3	DI	R	CO	M2
D6.1	Dissemination Plan and Website creation	WP6	COSBI	R	CO	M3
D2.1	Needs, requirements, and use cases	WP2	RO	R	PU	M6
D4.1	Report on core AI components and performance	WP4	COSBI	R	CO	M12
D6.2	Efficacy monitoring of communication to the general public via project website and social networks	WP6	COSBI	R	PU	M12
D1.2	Annual meeting year 1	WP1	COSBI	R	PU	M13
D3.2	Report on data collection	WP3	RO	R	CO	M13
D2.2	Development of a DT software framework	WP2	COSBI	OTHER	CO	M14
D5.2	Report on the ethical analysis	WP5	BH	R	PU	M20
D5.3	Report on patient's safety	WP5	BH	R	PU	M22
D1.3	Closing session documents and follow-up planning	WP1	COSBI	R	PU	M24
D2.3	DT user manual	WP2	COSBI	R	CO	M24

D4.2	Report on core AI components and performance	WP4	COSBI	OTHER	PU	M24
D5.1	Report on AI evaluation and DT user experience	WP5	COSBI	R	PU	M24
D6.3	Efficacy monitoring of communication to the general public via project website and social networks	WP6	COSBI	R	PU	M24

Table 3.1d: List of milestones (colors refer to different WP according to Figure 3)

Milestone (number)	Milestone name	Related work package(s)	Estimated date	Means of verification
MIL1	Project website	WP6	Month 3	Website running
MIL2	User needs and requirements	WP2	Month 6	Deliverable D2.1 released
MIL3	Data collection	WP3, WP5	Month 13	Availability of >90% of the data to be collected
MIL4	AI core components	WP4, WP5	Month 13	Release of the >75% of the source code of the AI core algorithms
MIL5	DT software infrastructure	WP2, WP5	Month 18	Release of the first version of the code to setup the DT
MIL6	Project end	WP1, WP6	Month 24	Acceptance of final report of the project

Table 3.1e: Critical risks for implementation (colors refer to different WP according to Figure 3)

Description of risk	Work package(s) involved	Proposed risk-mitigation measures
Management and dissemination	WP1, WP6	Low risk. We will organize regular meetings to monitor the developments of the project and workshops for the dissemination of the results
Failure to collect sufficient enough data for AI models	WP3, WP4	Low/medium risk. We explicit publicly available datasets and some private datasets already available at UCBM. 1 milestone and 1 deliverable are dedicated to checking and reporting data acquisition problems and their solutions.
One partner leaves the consortium reducing the technical expertise and/or the content provision	WP1, WP2, WP3, WP4, WP5, WP6	We can contact other institutions to get involved in the project exploiting the research networks of all the partners
National Covid-19 restrictions prevent or limit in-person co-design and evaluation studies	WP2, WP6	Low-medium risk. Studies will be rearranged to be delivered online where feasible or by other partners.
No consensus on the ethical approaches	WP1, WP3	Low risk. The consortium will initiate rigorous consultation with the Ethics Committee to resolve and address the lack of consensus on the ethical approach. If this approach still fails to provide an accepted solution, external consultants will be inquired with.
Dissemination and exploitation do not have planned impact or lack of interest for dissemination and exploitation plans (low-high)	WP6	Medium risk. Organize frequent interaction between industrial and academic partners.
DTs of the patients failed to reach the challenging objectives of the project	WP2, WP4, WP5	Medium risk. Quite all the partners are dedicated to DT design and implementation.

The developed DT is not accepted by the end-users for different reasons such as the difficulty of use, or the difficulties of the learning phase (low-medium)	WP2, WP5	During the software development of the platform, the analysis of the user needs will be taken into account. Furthermore, in the prototype development phase, a modular architecture will be set up, in which it will be easy to intervene to improve the software components and their integration during the validation.
Unsatisfying results on DT test	WP5	Medium risk, this possibility won't affect the diffusion of new datasets for future DT research. Moreover, preliminary results obtained in precision medicine by UCBM discourage this eventuality.

3.2 Resources to be committed

Table 3.3a: Summary of staff effort

	WP1	WP2	WP3	WP4	WP5	WP6	Total person/months per RU
RU n° 01/ COSBI	2	11	15	21	8	9	56
RU n 02 / RO	1	2	7	2	2	2	16
RU n° 03/ DI	1	2	6	1	2	2	14
RU n° 04/ BH	2	1	3	1	1	2	10
CDI	2	2	1	0	2	2	9
Total person/months	8	18	22	25	15	17	105

Table 3.3b: Other direct costs

RU 01/ COSBI	Cost (EUR)	Justification
<i>Travel</i>	1.000 €	Conference participation to present the results, participation to national and international meetings on the topics of the project
<i>Equipment</i>	0 €	No needed as the RU already have workstations and storage devices
<i>Other goods and services</i>	1.000 €	Press conference and other dissemination activities, open access publication (shared with other RUs)
<i>Total</i>	2.000 €	

RU 02/ RO	Cost (EUR)	Justification
<i>Travel</i>	1.000 €	Conference participation to present the results, participation to national and international meetings on the topics of the project
<i>Equipment</i>	0 €	No needed
<i>Other goods and services</i>	1.200 €	Press conference and other dissemination activities, open access publication (shared with other RUs)
<i>Total</i>	2.200 €	

RU 03/ DI	Cost (EUR)	Justification
<i>Travel</i>	1.000 €	Conference participation to present the results, participation to national and international meetings on the topics of the project
<i>Equipment</i>	0 €	No needed
<i>Other goods and services</i>	1.200 €	Press conference and other dissemination activities, open access publication (shared with other RUs)
<i>Total</i>	2.200 €	

RU 04/ BH	Cost (EUR)	Justification
<i>Travel</i>	1.000 €	Conference participation to present the results, participation to national and international meetings on the topics of the project
<i>Equipment</i>	0 €	No needed
<i>Other goods and services</i>	1.200 €	Press conference and other dissemination activities, open access publication (shared with other RUs)
<i>Total</i>	2.200 €	

4. Participants

Next table 4.1 shows the role of participating RUs in the project.

Table 4.1: Role of participating RUs

Role	COSBI	RO	DI	BH	CDI
Project management	☒	☒	☒	☒	☐
Communication, dissemination and engagement	☒	☒	☒	☒	☒
Provision of research and technology infrastructure	☒	☒	☒	☐	☐
Co-definition of research and market needs	☒	☒	☒	☒	☒
Healthcare representative	☐	☒	☒	☐	☒
Industrial representative	☐	☐	☐	☐	☒
Policy maker or regulator, incl. standardization body	☒	☐	☐	☒	☐
Research performer	☒	☒	☒	☒	☐
Technology developer	☒	☐	☐	☐	☐
Testing/validation of approaches and ideas	☒	☒	☒	☒	☒
Prototyping and demonstration	☒	☒	☒	☐	☒
Intellectual Property Rights management	☒	☐	☐	☒	☐
Public procurer of results	☒	☒	☒	☒	☐
Private buyer of results	☐	☐	☐	☐	☒
Finance or in-kind provider (public or private)	☐	☐	☐	☐	☒
Education and training	☒	☒	☒	☒	☐
Contributions from the social sciences or/and the humanities	☒	☒	☒	☒	☐

Unit of Computer Systems and Bioinformatics (COSBI): It deals with intelligent processing of data, signals, images and videos with particular reference to biomedical applications. The research activity has been focusing on developing new methods for AI, mostly focusing on machine and DL, also to respond to the new needs of the application areas. Indeed, with reference to the applications, they have impacted biomedical field, using unimodal and multimodal data collected from different sources. Some examples are: radiomics, precision medicine, COVID-19 prognosis, COPD prediction, etc.

COSBI profile perfectly matches with this project, which applies results already attained (table 1.3), and also pushes forward the research integrating DT with advanced AI techniques.

Researchers involved in the proposal

Participant	Gender	Nationality	E-mail	Career Stage	Role in the project
Prof. Paolo Soda	Male	Italian	p.soda@unicampus.it	Full Professor	PI
Paolo Soda graduated in Biomedical Engineering from University Campus Bio-Medico (UCBM) in Rome in 2004 and received his PhD in Biomedical Engineering (computer science area) in 2008 from the same University. He is Full Professor in Computer Science at UCBM, Visiting Professor in Artificial Intelligence and Biomedical Engineering at the Department of Radiation Sciences, Umeå University, Sweden, and he has been serving as Research and Third Mission Coordinator, at the Department of Engineering, UCBM. He is also the Deputy Coordinator of the National PhD program in Artificial Intelligence, Health and Life Sciences area. His research interests include AI, machine learning, big data analytics, data mining, 2D and 3D image and video processing and analysis. He is a member of programme committees of international conferences and has been guest editor of international impact journals. Since 2012, he has been an associate editor of the proceedings of the Annual Int Conference of the IEEE Engineering in Medicine & Biology Society. From 2018 to 2021, he chaired the steering committee of the IEEE Int Symposium of Computer-Based Medical Systems. He has been co-PI of national and international projects. His research activity is certified by >140 scientific publications, >1850 citations, h-index 24 (Scopus, April 18, 2023). He is co-author of 4 papers in award-winning international conference proceedings (IEEE LSC 2018, IEEE BIBM 2018, IEEE ICCI*CC 2019, IEEE CBMS 2021). He coordinated the winning working group of 2 international competitions: "COVID CXR Hackathon" (2022 Dubai Expo) and "All against COVID-19: Screening X-ray Images for COVID-19 Infection" (IEEE 2021). He is a member of IEEE, CVPL and SIBIM; from 2017 to 2022 he chaired the IEEE International Technical Committee for Computational Life Sciences. He is co-founder of BPCOmedia srl.					
Dr. Valerio Guarrasi	Male	Italian	valerio.guarrasi@unicampus.it	Post-Doc	Participant
Postdoctoral Researcher at the Campus Bio-Medico University of Rome. He received the Ph.D. in Data Science at Sapienza University of Rome in 2023. His research interests are in DL, pattern recognition and machine learning. He has been working extensively on MDL integrated with XAI methods on many bio-medical applications.					
Eng. Camillo Maria Caruso	Male	Italian	camillomaria.caruso@unicampus.it	Ph.D. Student	Participant
Ph.D. student at Campus Bio-Medico University of Rome in the Italian National PhD Program in Artificial Intelligence. He received her Master's degree at the same University in 2020. His research interests are DL, pattern recognition and radiomics.					
Eng. Aurora Rofena	Female	Italian	aurora.rofena@unicampus.it	Ph.D. Student	Participant
Ph.D. student at Campus Bio-Medico University of Rome in the Italian National PhD Program in Artificial Intelligence. She received her Master's degree at the same University in 2023. Her research interests are MDL and generative AI.					
Relevant publications					
Caruso, C. M., Guarrasi, V., Cordelli, E., Sicilia, R., Gentile, S., Messina, L., ... & Soda, P. (2022). A					

Multimodal Ensemble Driven by Multiobjective Optimisation to Predict Overall Survival in Non-Small-Cell Lung Cancer. <i>Journal of Imaging</i> , 8(11), 298.
Soda, P., D'Amico, N. C., Tessadori, J., Valbusa, G., Guarrasi, V., Bortolotto, C., ... & Papa, S. (2021). AIforCOVID: Predicting the clinical outcomes in patients with COVID-19 applying AI to chest-X-rays. An Italian multicentre study. <i>Medical image analysis</i> , 74, 102216.
Ramella, S., Fiore, M., Greco, C., Cordelli, E., Sicilia, R., Merone, M., ... & Soda, P. (2018). A radiomic approach for adaptive radiotherapy in non-small cell lung cancer patients. <i>PloS one</i> , 13(11), e0207455.
Castiglioni, I., Gallivanone, F., Soda, P., Avanzo, ... & Salvatore, M. (2019). AI-based applications in hybrid imaging: how to build smart and truly multi-parametric decision models for radiomics. <i>European Journal of Nuclear Medicine and Molecular Imaging</i> , 46, 2673-2699.
Foggia, P., Percannella, G., Soda, P., & Vento, M. (2013). Benchmarking HEp-2 cells classification methods. <i>IEEE transactions on medical imaging</i> , 32(10), 1878-1889.
Relevant previous projects or activities
<i>FAIR: Future AI Research</i> , supported by PNRR, Mission 4 Component 2. The goal is to advance frontier research in AI, reduce AI research fragmentation, create human-centered, robust, trustworthy and sustainable AI, foster AI-based innovation and development of AI technology, attract AI talents. Duration: 2023-2026
<i>Platform for Precision Medicine. Artificial Intelligence and Integrated Clinical Diagnostics</i> , supported by FONDO PER LA CRESCITA SOSTENIBILE (F.C.S.), Bando Accordo Innovazione DM 24/5/2017 (Ministero delle Imprese e del Made in Italy). Duration: 2022-2024
<i>CLARO: a CoLLaborative multi-sources Radiopathomics approach for personalized Oncology in non-small cell lung cancer</i> : supported by UCBM in the 2018 internal call. The project reached promising results and was positively evaluated by the reviews panel. Duration: 2018-2021
<i>AIforCOVID</i> : a national initiative enclosing several institutions in Italy, e.g., hospitals, research centers, which developed a unique open access dataset of images and clinical data to predict the outcome of COVID-19 patients. CDI partners here, participated to this initiative.
<i>Trustworthy AI-driven next generation precision medicine in COVID-19</i> : a project supported by Ministero degli Affari Esteri e Cooperazione Internazionale, with Shenzhen University. It deals with MDL and XAI. Duration: 2023-2025
Significant infrastructure and any major items of technical equipment
20000 hour/month GPU time: Grant supported by the Swedish National Infrastructure for Computing (SNIC), partially funded by the Swedish Research Council through grant agreement no. 2018-05973
The lab is equipped with several desktop computers that can be used for coding. Two workstations are also available (n. 2 INTEL XEON 5450 3.00 12M/1333 Q, 32 GB RAM, Nvidia GeForce GTX TITAN 6GB), and another workstation is available as Faculty core facility (n. 1 Intel Xeon E5-2697 v2 2.70Ghz 12Core, 192 GB RAM, NVIDIA TESLA K20c 5GB e NVIDIA Quadro K5000 4GB). Licenses of the main software are available.
The lab is acquiring two new workstations for DL, equipped with: i) 2 CPU AMD 64 CORE, 1 TB di RAM, 15 TB SSD, 4 GPU NVIDIA A100 ii) 2 CPU AMD 48 CORE, 1 TB di RAM, 15 TB SSD, 4 GPU A40
The lab is equipped with Network Attached Storage for the storage of big data: QNAP TS-451 with 4 bays, 2 Ethernet and 4 USB connections, and 1 HDMI port. It has 8GB of memory and a 2.0GHz quad-core Intel Celeron processor (64-bit), equipped with 2 Western Digital Red Pro disks of 12TB 3.5.

Unit of Radiation Oncology (RO): Has the latest equipment available for high precision radiotherapy. High precision radiotherapy allows for excellent tumor targeting and maximum sparing of normal tissue. Also in the division there is great interest in developing ways of increasing the effectiveness of radiotherapy by combining it with other treatments – including chemotherapy agents, targeted drugs or immunotherapies. A major research focus is the development of multimodality imaging for evaluating response to treatment in vivo.

The Research Unit of Radiation Oncology is the coordinator of this project. Its role in this project is: 1) selecting patients affected by NSCLC according to eligibility criteria, 2) treating stage III patients with chemoradiation, 3) manually contouring interest volumes in order to compute radiomic features; 4) collecting data on response, chemoradiation related toxicity and outcomes.

Researchers involved in the proposal

Participant	Gender	Nationality	E-mail	Career Stage	Role in the project
Prof. Sara Ramella	Female	Italian	s.ramella@policlinocampus.it	Full Professor	RU Key person
Full Professor in Radiation Oncology at the Campus Bio-Medico University in Rome; Member of Senate Academic. Her field of interest is clinical application of novel radiotherapy techniques such as Adaptive Radiotherapy, SABR (Stereotactic Ablative Radiotherapy), VMAT (Volumetric Arc Therapy) and IGRT (Image-Guided Radiation Therapy). She is actively involved in clinical and translational research. Current fields of investigation are radiomic, a tool to prognosticate and predict tumor response to oncological therapy, and LC treatment. She is recognized as an international expert in the treatment of non-small and small cell LC and in which way novel techniques can optimize such treatment. She is delegate of several national and european societies such as ESTRO, EORTC, IASLC, and was an invited speaker in more than eighty national and international meetings. She is reviewer for journals including Journal of International Journal of Radiation Oncology Biology and Physics, Radiotherapy and Oncology, Journal of Thoracic Oncology, Lung Cancer, Radiation Oncology, Clinical Lung Cancer, Journal of Proteomics & Bioinformatics, Breast Cancer Research and Treatment, Reviews on Recent Clinical Trials, Medical Dosimetry, BioMed Research International, Plos One, Journal of Supportive Care in Cancer and others. Professor Ramella has published original articles in national and international peer-reviewed journals.					
Prof. Michele Fiore	Male	Italian	m.fiore@policlinocampus.it	Associate Professor	Participant
Associate Professor at the Radiotherapy Research Unit of “Campus Bio-Medico” University of Rome. Education: in 2018 Assistant professor; in 2009 Certified in Radiation Oncology; in 2005 graduated in Medicine at Catholic University of Sacred Heart in Rome. He has experience as radiation oncologist in LC treatments and he is currently investigator of several research collaborations with other Universities about this topic. He is author of original articles in national and international peer-reviewed journals.					
Prof. Edy Ippolito	Female	Italian	e.ippolito@policlinocampus.it	Assistant Professor	Participant
Assistant Professor at the Radiotherapy Research Unit of “Campus Bio-Medico” University of Rome. Education: in 2018 Assistant professor; in 2008 Certified in Radiation Oncology; in 2004 graduated in Medicine at Catholic University of Sacred Heart in Rome. She has experience as radiation oncologist in LC segmentation, high technology radiotherapy treatments. She is author of original articles in national and international peer-reviewed journals.					
Dr. Carlo Greco	Male	Italian	c.greco@policlinocampus.it	Researcher	Participant
Researcher in Radiation Oncologist at the Radiotherapy Oncology Department of “Campus Bio-Medico” in Rome. Education: in 2011 Certified in Radiation Oncology; in 2006 graduated in Medicine at Campus Bio-Medico University in Rome. He is involved in clinical and translational research regarding LC treatments. He also has experience on use of lung functional imaging for radiotherapy treatments. He is author of original articles in national and international peer-reviewed journals.					

Relevant publications
Tortora M, Cordelli E, Sicilia R, Miele M, Matteucci P, Iannello G, Ramella S, Soda P. Deep Reinforcement Learning for Fractionated Radiotherapy in Non-Small Cell Lung Carcinoma. <i>Artif Intell Med.</i> 2021 Sep;119:102137. doi: 10.1016/j.artmed.2021.102137. Epub 2021 Aug 15. PMID: 34531006.
Sicilia R, et al (2018). Exploratory Radiomics for Predicting Adaptive Radiotherapy in Non-Small Cell Lung Cancer. In: <i>Computer-Based Medical Systems (CBMS), 2018 31st IEEE International Symposium on.</i> p. 250-255
Nestle U, et al. ESTRO ACROP guidelines for target volume definition in the treatment of locally advanced non-small cell lung cancer. <i>Radiother Oncol.</i> 2018 Apr;127(1):1-5. doi: 10.1016/j.radonc.2018.02.023. Epub 2018 Mar 28.
Ramella S, et al. Local Control and Toxicity of Adaptive Radiotherapy Using Weekly CT Imaging: Results from the LARTIA Trial in Stage III NSCLC. <i>J Thorac Oncol.</i> 2017 Jul;12(7):1122-1130. doi: 10.1016/j.jtho.2017.03.025. Epub 2017 Apr 17.
Warner A, et al. Factors Associated With Early Mortality in Patients Treated With Concurrent Chemoradiation Therapy for Locally Advanced Non-Small Cell Lung Cancer. <i>Int J Radiat Oncol Biol Phys.</i> 2016 Mar 1;94(3):612-20. doi: 10.1016/j.ijrobp.2015.11.030.
Relevant previous projects or activities
Identification of a radiomic signature to predict early response rate to chemoradiation in stage III lung cancer (NCI identifier trial number: NCT03583723)
Collaborative Laboratory of Precision Medicine & BioData Analytics National, collaboration of UCBM with Centro Diagnostico Italiano spa (CDI) in Milan involving COSBI laboratory and the Radiation Oncology unit.
Convention on radiomics, “Radiomica: il futuro è qui”, Milan, Italy, October 28, 2018, in cooperation with CDI.
Convention on radiomics, “Radiomica: il futuro è qui”, Naples, Italy, November 19, 2018, in cooperation with CDI.
Significant infrastructure and any major items of technical equipment
The department of Radiation Oncology of Campus Bio-Medico University has an high technology equipment like Intensity Modulated Radiotherapy (IMRT, including volumetric dynamic arc IMRT-VMAT), Image-Guided Radiotherapy (IGRT), respiratory gating, intra- and extra-cranial stereotactic radiotherapy and 3-D conformal radiotherapy. The department treats over 1500 patients each year and includes 4 linear accelerators on two different locations and a day hospital for concurrent oncological treatments.

Unit of Diagnostic Imaging (DI): in this Unit are carried out instrumental examinations, using different techniques, such as conventional radiology (x-rays), mammography, ultrasonography (ultrasound), bone densitometry, angiography, magnetic resonance imaging (MRI) and multislice computed tomography (CT), aiming to provide useful information on the normal/abnormal morphology, perfusion and function of different organs and systems. One of the major aims of the Unit is to accelerate the instrumental evaluation of the patients, to interpret the images and to refer, as soon as possible, about the results of different examinations to help the planning of the therapy. Another important role is the monitoring of the effect of different therapies in oncologic patients to adapt and personalize the treatment to reach the maximum therapeutic efficacy. Its role in this project is to contribute to create patient-specific biophysical model, using all imaging examinations of the patient, to provide visual 3D organ reconstructions, that integrated with other kind of medical data (history, laboratory, signs, symptoms etc. etc.), are the basis where simulating various treatment scenarios enabling the medical team to select the optimal one.

Researchers involved in the proposal

Participant	Gender	Nationality	E-mail	Career Stage	Role in the project
Prof. Bruno Beomonte Zobel	Male	Italian	b.zobel@unica mpus.it	Full Professor	RU Key person
Full Professor in Diagnostic Imaging at the Campus Bio-Medico University in Rome; Head of the Imaging Center and Coordinator of the Diagnostic Imaging and Interventional Radiology research unit. His field of interest are research and innovation in diagnosing, planning and monitoring treatment and predicting outcome in Oncology. Technology advances in digital imaging have enabled the images produced to be post-processed, enhanced and used for planning radiotherapy, surgical or interventional treatments in oncology patients. He is actively involved in informatics applications in medical research. Current fields of investigation are artificial intelligence, radiomic, augmented reality applications in Diagnostic Imaging and Interventional Radiology. He is recognized as an international expert in the use of artificial intelligence algorithms to support radiological activities. He is in the board of directors of the Italian Society of Medical and Interventional Radiology (SIRM) and was an invited speaker in more than two hundred national and international scientific meetings. Professor Beomonte Zobel was the medical Principal Investigator in different research projects co-founded by the European Commission and published more than four hundred original scientific articles in national and international peer-reviewed journals.					
Prof. Rosario Francesco Grasso	Male	Italian	R.Grasso@poli clinicocampus .it	Assistant Professor	Participant
Assistant Professor in Diagnostic Imaging at the Campus Bio-Medico University in Rome; Head of the Interventional Radiology unit. His fields of interest are percutaneous biopsies, thermal ablations and cryoablations of solid tumors, especially of lung cancers. He is recognized as an international expert in the use of augmented reality in guiding interventional oncology. Professor Grasso published more than one hundred original scientific articles in national and international peer-reviewed journals and was an invited speaker in several tens national and international scientific meetings.					
Dr. Carlo Augusto Mallio	Male	Italian	C.Mallio@poli clinicocampus. it	Researcher	Participant
Assistant Professor in Neuroradiology at the Campus Bio-Medico University in Rome. He obtained the Medical Degree in 2009 and the Board Certification in Radiology in 2015, both at the Campus Bio-Medico University in Rome. Then, he obtained the PhD degree in Medical Sciences in 2021 at the University of Antwerp and a II Level Master Degree in Diagnostic Neuroradiology in 2023 at Sapienza Università di Roma. Current fields of research include Oncologic Imaging, Neuroimaging, Imaging Processing, and Artificial Intelligence applications in Radiology.					

Relevant publications
Mallio CA, Radbruch A, Deike-Hofmann K, Van der Molen AJ, Dekkers IA, Zaharchuk G, Parizel PM, Beomonte Zobel B, Quattrocchi CC. Artificial Intelligence to Reduce or Eliminate the Need for Gadolinium-Based Contrast Agents in Brain and Cardiac MRI: A Literature Review. Invest Radiol. 2023 (In Press).
Greco F, Mallio CA. Artificial intelligence and abdominal adipose tissue analysis: a literature review. Quant Imaging Med Surg. 2021 Oct;11(10):4461-4474. doi: 10.21037/qims-21-370. PMID: 34603998; PMCID: PMC8408793.
Mallio CA, Quattrocchi CC, Beomonte Zobel B, Parizel PM. Artificial intelligence, chest radiographs, and radiology trainees: a powerful combination to enhance the future of radiologists? Quant Imaging Med Surg. 2021 May;11(5):2204-2207. doi: 10.21037/qims-20-1306. PMID: 33937001; PMCID: PMC8047344.
Mallio CA, Napolitano A, Castiello G, Giordano FM, D'Alessio P, Iozzino M, Sun Y, Angeletti S, Russano M, Santini D, Tonini G, Zobel BB, Vincenzi B, Quattrocchi CC. Deep Learning Algorithm Trained with COVID-19 Pneumonia Also Identifies Immune Checkpoint Inhibitor Therapy-Related Pneumonitis. Cancers (Basel). 2021 Feb 6;13(4):652. doi: 10.3390/cancers13040652. PMID: 33562011; PMCID: PMC7914551.
Quattrocchi CC, Mallio CA, Presti G, Beomonte Zobel B, Cardinale J, Iozzino M, Della Sala SW. The challenge of COVID-19 low disease prevalence for artificial intelligence models: report of 1,610 patients. Quant Imaging Med Surg. 2020 Sep;10(9):1891-1893. doi: 10.21037/qims-20-782. PMID: 32879867; PMCID: PMC7417757.
Relevant previous projects or activities
<i>CLARO: a CoLLaborative multi-sources Radiopathomics approach for personalized Oncology in non-small cell lung cancer</i> : supported by UCBM in the 2018 internal call. The project reached promising results and was positively evaluated by the reviews panel. Duration: 2018-2021
<i>Platform for Precision Medicine. Artificial Intelligence and Integrated Clinical Diagnostics</i> , supported by FONDO PER LA CRESCITA SOSTENIBILE (F.C.S.), Bando Accordo Innovazione DM 24/5/2017 (Ministero delle Imprese e del Made in Italy). Duration: 2022-2024
Significant infrastructure and any major items of technical equipment
In the Diagnostic Unit instrumental examinations are carried out, using different techniques, such as conventional radiology (x-rays), mammography, ultrasonography (ultrasound), bone densitometry, angiography, magnetic resonance and multislice computed tomography (CT), aiming to provide useful information on the normal/abnormal morphology, perfusion and function of different organs and systems of human body. The Diagnostic Imaging Unit is fit out with the following instrumental equipments: four computed tomography systems, one of them with a dual energy technology and 256 rows of detectors; three magnetic resonance tomographies, operating at 1,5 Tesla; two mammographic units; four ultrasound machines; two angiographic systems; five digital radiology units. The department equipments are connected with a high speed network to twenty five desktop computers, using each two high resolution screens, and to a radiology information system and a picture archiving and communication system, fully integrated between them.

Unit of Bioethics and Humanities (BH): carries on its activity thanks to a multidisciplinary team (medical doctors, philosophers, psychologists, nurses, and legal experts). This teamwork makes it possible to detect when research and testing are really placed at the service of mankind, with respect to the Ethics of Job Well Done. The fields of research include Forensic and Legal Medicine (Professional liability, Clinical Risk Management, Quality and Innovation of Care, etc.), General Bioethics (Theory of the Ethics of Job Well Done), Clinical Bioethics (Clinical Method, End-of-life treatment, Clinical Psychology), Applied Ethics (Medicine, Engineering, and Human Nutrition), Moral Philosophy (Theory of Anthropology and General Ethics) and Medical Humanities. Project activities, conferences, conventions, lessons and publicist activity are carried out in partnership with qualified national and international research groups.

Researchers involved in the proposal

Participant	Gender	Nationality	E-mail	Career Stage	Role in the project
Dr. Francesco De Micco	Male	Italian	f.demicco@poli clinicocampus.it	Researcher	RU Key person

Researcher in Legal Medicine at the Campus Bio-Medico University in Rome. He is a specialist in Legal Medicine, PhD in Translational and Clinical Medicine, MSc in Occupational Medicine and MSc in Social Security and Social Insurance Medicine. He has been qualified as an associate professor for the subject area 06/M2 - Legal and Labour Medicine. He holds the following positions at Campus Bio-Medico University Hospital: Clinical Risk Manager, President of the Claims Assessment Committee, Member of the Clinical Bioethics Service, Member of the Committee for the Control of Infections Related to Assistance. He is member of the Bioethics and Humanities Research Unit at Campus Bio-Medico University in Rome. His research interests are mainly in the field of clinical risk management, patient safety, quality of care, bioethics of a "job well done" in public health, medical liability, personal injury evaluations, use of diagnostic imaging in forensic anthropology. He has participated as a speaker or organizer of national and international conferences and conventions. Academic Editor PLOS ONE, Topic Editor Special Issue "Equitable Digital Medicine and Home Health Care", Co-Editor peer-reviewed Edited Volume "Forensic and Legal Medicine - State of the Art, Practical Applications and New Perspectives", Editor Special Issue "Best Practice: Proposals for Preparing Pandemics Governance, Topic Editor Special Issue "Ethics and COVID-19: The Bioethics of a "Job Well Done" in Public Health", Editorial Board Member of Forensic Sciences. He is Associate Member of the "Società Italiana di Medicina Legale e delle Assicurazioni", "International Academy of Legal Medicine", "Italian Network for Safety in Healthcare", "American Academy of Forensic Science", "Gruppo Italiano Antropologi e Odontologi Forensi", "Società Italiana di Odontoiatria Forense" and "Gruppo Italiano di Patologia Forense". Dr. De Micco has published original articles in national and international peer-reviewed journals.

Relevant publications

Travaini GV, Pacchioni F, Bellumore S, Bosia M, De Micco F. Machine Learning and Criminal Justice: A Systematic Review of Advanced Methodology for Recidivism Risk Prediction. International Journal of Environmental Research and Public Health. 2022; 19(17):10594. <https://doi.org/10.3390/ijerph191710594>

De Micco F, Fineschi V, Banfi G, Frati P, Oliva A, Travaini GV, Picozzi M, Curcio G, Pecchia L, Petitti T, Alloni R, Rosati E, De Benedictis A, Tambone V. From COVID-19 Pandemic to Patient Safety: A New "Spring" for Telemedicine or a Boomerang Effect? Front Med (Lausanne). 2022 Jun 15;9:901788. doi: 10.3389/fmed.2022.901788. PMID: 35783642; PMCID: PMC9240206.

Tambone V, Frati P, De Micco F, Ghilardi G, Fineschi V. How to fix democracy to fix health care. Lancet. 2022 Jan 29;399(10323):433-434. doi: 10.1016/S0140-6736(21)02802-6. PMID: 35093222; PMCID: PMC8797029.

De Micco F, De Benedictis A, Fineschi V, Frati P, Ciccozzi M, Pecchia L, Alloni R, Petrosillo N, Filippi S, Ghilardi G, Campanozzi LL, Tambone V. From Syndemic Lesson after COVID-19 Pandemic to a "Systemic Clinical Risk Management" Proposal in the Perspective of the Ethics of Job Well Done. International Journal of Environmental Research and Public Health. 2022; 19(1):15. <https://doi.org/10.3390/ijerph19010015>

De Micco F, Martino F, Campobasso CP. Ethical issues in age assessment by the third molar development. Australian Journal of Forensic Sciences, DOI: 10.1080/00450618.2020.1789220

Relevant previous projects or activities

RELA-Verso: Use of the metaverse to prevent aggression against health workers. The main objective is to realize a virtual reality/augmented reality to foster the ability of HWs to deal with highly emotional situations and to disseminate to all staff the knowledge of potential risks and techniques to be followed to protect themselves and their colleagues from acts of violence.

DAYCARE (Design and Implementation of a Systemic Clinical Risk Management): the objective of the Think Tank is to propose an innovative clinical risk management model for improving the quality of care and patient safety through the interpretation and correlation of Big Data using new technologies such as AI, IoT and Robotics.

AgEstimation project: international research project on age estimation through medical imaging.

Pandemic preparedness and response: In vitro diagnostic devices to tackle cross-border health threats. This project proposes the development, preclinical and clinical validation and commercialization of a Point of Care biosensing platform based on multiplexed field-effect sensor technology based on graphene monolayers functionalized with specific and oriented recognizing biomolecules.

4.1 Third parties involved in the project (including use of third party resources)

Does the participant plan to subcontract certain tasks (please note that core tasks of the project should not be sub-contracted because research funding is supposed to support RUs of the university only)	No
Does the participant envisage that part of its work is performed by linked third parties? <i>A third party is an affiliated entity or has a legal link to a participant implying a collaboration not limited to the action.</i>	Yes
Centro Diagnostico SpA endorse this project and offer in-kind support. It is involved in WP1 (T1.1, T1.2, T1.3, T1.5), WP2 (T2.1, T2.2), WP3 (T3.2, T3.3, T3.4), WP5 (T5.1, T5.2), WP6 (T6.5). Their involvement in the project guarantees possible exploitation beyond the project itself and can also offer useful insights when defining the needs and requirements. Nevertheless, the project is organized in a way so that its goals can be fully accomplished even without their participation.	

5. Ethics and security

5.1 Ethics

ETHICAL ISSUES TABLE	YES/NO	Reference page of the proposal
Informed Consent		
Does the proposal involve children?	NO	
Does the proposal involve patients or persons not able to give consent?	NO	
Does the proposal involve adult healthy volunteers?	NO	
Does the proposal involve Human Genetic Material?	NO	
Does the proposal involve Human biological samples?	NO	
Does the proposal involve Human data collection?	YES	14, 15, 26, 17
Does your research involve physical interventions on the study participants?	NO	
Does it involve invasive techniques?	NO	
Research on Human embryo/foetus		
Does the proposal involve Human Embryos?	NO	
Does the proposal involve Human Foetal Tissue / Cells?	NO	
Does the proposal involve Human Embryonic Stem Cells?	NO	
Privacy		
Does the proposal involve processing of genetic information or personal data (eg health, sexual lifestyle, ethnicity, political opinion, religious or philosophical conviction)	YES	14, 15, 26, 17
Does the proposal involve tracking the location or observation of people?	NO	
Research on Animals		
Does the proposal involve research on animals?	NO	
Are those animals transgenic small laboratory animals?	NO	
Are those animals transgenic farm animals?	NO	
Are those animals cloning farm animals?	NO	
Are those animals non-human primates?	NO	
Research Involving Developing Countries		
Use of local resources (genetic, animal, plant, etc)	NO	
Benefit to local community (capacity building ie access to healthcare, education etc)	NO	
Dual Use		
Research having potential military / terrorist application	NO	
Research having the potential for terrorist abuse	NO	

This proposal is in line with the inspiring principles of the UCBM, as it aims at providing more effective, efficient, safe and sustainable care to people affected by NSCLC. Furthermore, it brings together professionals with diverse backgrounds and expertise, in the spirit of sharing knowledge and efforts within a common project.

Given the involvement of human subjects in this study, all retrospective patient data will be retrieved according to a protocol already approved by our Ethical Committee (registration number 16/19 OSS ComEt CBM). Furthermore, in April 2023 we submitted a request of emendation to such protocol in order to retrieve the data needed for this project which was not previously included. Please find attached the approved protocol and emendation.

Sensitive data will be stored in compliance with the current regulation on privacy and data protection. This project is not expected to have any impact on any issue with ethical relevance.

5.2 Security

Activities or results raising security issues: NO